Regularized Masked Auto-Encoder for Semi-Supervised Hyperspectral Image Classification

Liguo Wang, *Member, IEEE*, Heng Wang, Peng Wang, *Senior Member, IEEE*, and Lifeng Wang

***Abstract*— As the most prevalent self-supervised representation learning (SSRL) model, the masked auto-encoder (MAE) has been gradually investigated in semi-supervised hyperspectral image classification (SHIC). However, the majority of the current approaches augment MAE merely from the application perspective or by introducing a weak regularization term, and do not comprehensively consider the challenges posed by the high intraclass variances and interclass similarities that often appear in hyperspectral image (HSI) data. In this article, we present a regularized MAE (RMAE) to address the aforementioned problems. Specifically, within the framework of MAE, we introduce a self-designed induced transformer block, using a small number of visible patches to learn the embeddings of patches with larger receptive fields. The learned embeddings are used to reconstruct the corresponding patches and an induced reconstruction loss is calculated. This strategy creates a much harder task for masked image modeling (MIM), and the induced transformer block is lightweight and imposes negligible computational burden overhead the underlying MAE framework. In addition, by rethinking the masking operations, we develop a masked convolutional neural network (MCNN), uncovering the principle of MAE and affirming the efficacy of RMAE. Finally, we present two metrics: the mean intraclass distance, and the mean interclass distance. Based on the metrics we give two criteria to evaluate the performance of an SSRL model, providing a new coordinate for the research in SSRL-based SHIC. Experiments conducted on four publicly accessible datasets show that RMAE outperforms state-of-the-art methods. The source code was powered by Jupyter and released at https://github.com/swiftest/RMAE.**

***Index Terms*— Induced transformer, masked image modeling (MIM), regularized masked auto-encoder (MAE), self-supervised representation learning (SSRL), semi-supervised hyperspectral image classification (SHIC).**

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I. Introduction

HYPERSPECTRAL images (HSIs) are inherently rich in spectral–spatial information. As a basic task, HSI classification (HIC) plays an indispensable role in the fields of agricultural monitoring, mineral exploration, environmental supervision, and other related domains. The rapid advancement of deep learning offers a steady stream of new technical support for HIC [1], [2], [3], [4]. It is noteworthy that the problem of dimensional curse caused by spectral redundancy is no longer a concern that must be emphasized in the face of data-hungry deep models, as the models often possess considerable model capacity, enabling them to accommodate the complete spectral information [5], [6], [7], [8].

The field has witnessed the emergence of several outstanding fully supervised classification models. Wang et al. [9] proposed a dual context model (DCM) based on transformer encoders and decoders, which uses transformer [10] to effectively encodes regional adaptation and global aggregation spatial context within and between homogeneous regions discovered by clustering, and then uses a convolution layer and a bilinear up-sampling layer for predictions. To learn multiscale neighborhood spatial context, Yang et al. [11] introduced an interactive learning framework called center-to-surrounding interactive learning (CSIL). This framework enables multiscale, detail-aware, and spatial interactive classification based on a meticulously designed hierarchical region sampling strategy, center transformer, and surrounding transformer. Sun et al. [12] designed a heterogeneous spectral–spatial network with 3-D attention and multilayer perceptron (HSSAM). The network embeds SimAM (simple attention module) into a 3-D multiscale residual-dense asymmetric convolution module to fully extract spectral–spatial features while avoiding sightless spots and unrecognized regions within the receptive field. Although HSSAM has been proven to be able to obtain high-precision classification results, excessive 3-D convolution operations lead to a substantial increase in model complexity. Roy et al. [13] used 3-D convolution kernels with different spatial sizes to mine adaptive spectral–spatial joint features, and combined various residual blocks to construct an attention-based adaptive kernel residual network (A2S2K-RN). Ravikumar et al. [14] applied the deep matrix capsule network (DMCN) [15] to HIC. The most special thing about DMCN is that it is based on the core idea of viewpoint invariant parse tree, learning the pose and activation

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probability of each capsule feature in different capsule layers. Hong et al. [16] designed SpectralFormer, which uses several consecutive patch-wise spectral bands as input to the transformer and uses a cross-layer adaptive fusion (CAF) module to fuse the outputs of different transformer blocks to enhance information exchange. Sun et al. [17] invented a memory-augmented spectral–spatial transformer (MASSFormer) for HIC, which applies maximum pooling and average pooling to the output of the convolutional module to obtain the memory tokens, and cascades the memory tokens to the end of keys and values in each transformer encoder to retain memory. While MASSFormer is effective, the fact that each transformer block uses the same fixed memory tokens may limit the performance of the model somewhat. Chen et al. [18] revisited the role of the center pixel in a transformer and put forward a spectral query spatial transformer (SQSFormer) framework. SQSFormer integrates random central rotation and center relative position embedding to mitigate the interference of absolute position information on spatial feature extraction, and uses multiple stacked central pixel embeddings to replace the class token to accomplish the classification task. In our previous study [19], we constructed a capsule vision transformer (CapsViT) for HIC. CapsViT acquires global correlations among various primary capsules by introducing a vision transformer (ViT) [20] into the capsule network. Both DMCN and CapsViT are appealing, nevertheless, their training speed is sluggish because of the existence of the iterative routing mechanism. Definitely, the remote sensing community also encompasses lightweight classification models, such as spectral–spatial feature tokenization transformer (SSFTT) proposed by Sun et al. [21], fast patch-free global learning (FPGA) proposed by Zheng et al. [22], and hybrid spectral net (HybridSN) proposed by Roy et al. [23]. In general, the fully supervised classification models aim to extract distinguishable spectral–spatial joint features, and make the intraclass embedding distance smaller and the interclass embedding distance larger as much as possible.

However, the labeling of samples in practical applications presents a distinct challenge due to its time-consuming and labor-intensive nature. At the same time, the potential learnable properties of tons of unlabeled samples are not fully utilized. For the reasons, semi-supervised HIC (SHIC) has been an active research topic [24], [25], [26], [27]. Liu et al. [28] used Ladder (an auto-encoder with lateral skip connections) network to integrate unsupervised feature learning and supervised classification into a single framework, and experimentally verified that the unsupervised learning objective of the network can greatly improve the classification accuracy. Jia et al. [29] proposed a novel semi-supervised Siamese network (3DAES, 3-D auto-encoder Siamese). The network incorporates a 3-D auto-encoder to obtain the underlying deep features of all samples (both labeled and unlabeled), and employs a Siamese network to correct the unsupervised features, thereby reducing the intraclass distance and increasing the interclass distance. While the simplistic 3-D auto-encoder is capable of learning the refinement representations and creating unsupervised features, the simplistic encoded features it acquires, even when corrected by the Siamese network, make a negligible contribution to the ultimate classification.

In addition to semi-supervised classification methods based on auto-encoders, methods based on pseudo labels are also widely concerned. Fang et al. [30] presented a collaborative learning framework that first trains a lightweight 3-D convolutional neural network (CNN) based on depth-wise and point-wise convolutions (3D-DPNet) with labeled samples, and then uses approximate rank-order clustering (AROC) to cluster the features of all samples extracted from the pretrained 3D-DPNet. At last, the predicted labels of the unlabeled samples are compared with the clustering labels decided by majority vote, and the training set is extended with the samples with the same label, and the iterative training is carried out. Seydgar et al. [31] developed a probabilistic pseudo-label generation framework for SHIC, which assumes that the coding features of labeled samples follow a Gaussian distribution, and then estimates the distribution using maximum likelihood estimation (MLE). When assigning labels to unlabeled samples, the framework uses a confidence interval to measure the similarity score between the encoded features of the unlabeled samples and the distribution of the labeled samples, and generates false labels for samples with high confidence. Mukherjee and Prasad [32] proposed a new spectral–spatial semi-supervised deep learning framework based on angular discriminant loss. The framework first applies the constrained Dirichlet process mixture model (C-DPMM) to cluster the original HSI data to generate cluster labels, and then uses the cluster labels to build training sample pairs. A Siamese network is trained, with its loss function encompassing categorical cross-entropy and angularly discriminative loss. Finally, the trained Siamese network is used for supervised tuning. Wu and Prasad [33] introduced a classical pseudo-label-based semi-supervised classification method known as semi-supervised deep learning using pseudo labels (PL-SSDL), which uses C-DPMM to cluster the input HSI, and uses limited labeled samples for tuning after pretraining with false labels. Although PL-SSDL obtains satisfactory classification results, the complexity of the method is unacceptable to some extent. Cao et al. [34] put forward a semi-supervised feature learning approach for disjoint HIC (SFDC), which uses label propagation to make pseudo-label predictions on unlabeled samples, and alternately iterates the two steps of label prediction and feature optimization until the label prediction becomes stable. Feng et al. [35] proposed a new semi-supervised learning framework for unbalanced HIC, which proposes a strategy for reconfirming candidate sample labels used to expand the training set based on the label information of spatial neighborhood samples. At the same time, to solve the class imbalance problem, the framework uses an interclass data rebalancing strategy, that is, the lower the frequency of one category of samples in the initial training sample, the higher the frequency of the selection of the same pseudo-label samples. Xu et al. [36] put forward a robust self-ensembling network (RSEN), in which the base network uses labeled samples to calculate the cross-entropy loss, and unlabeled samples are fed into both the base network and the ensemble network to calculate the mean square error

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of the predicted outputs of both networks. The base network is updated using gradients from both losses, while the ensemble network is manually updated using the exponential moving average of historical parameters in the base network. In addition, to ensure the quality of the generated pseudo labels, RSNE adopts a consistency filtering strategy to retain unlabeled samples with a high degree of consistency in the predictions of the ensemble network and expand the training set. Admittedly, semi-supervised classification approaches based on pseudo labels are indeed simple and straightforward, however, false labels can cause more or less damage to the classifier.

To avoid the influence of false labels on classification models, graph-based SHIC methods have been widely studied [37], [38], [39]. To better capture the global graph structure and extract valuable local information, Ding et al. [40] proposed a new semi-supervised locally preserving dense graph neural network (GNN) with autoregressive moving average (ARMA) filter and context-aware learning (DARMA-CAL), which effectively tackles the issue of over-smoothing in current graph convolutional networks (GCNs) as well. Sha et al. [41] and Zhao et al. [42] explored the spectral–spatial graph attention network for SHIC, the core idea of which is to accumulate the neighborhood features to the current node in the multilayer propagation rules of GCNs, so as to make full use of the topological information of the graph structure. Considering that existing classification models based on GCNs cannot understand the global context information of the graph, Ding et al. [43] proposed a semi-supervised network based on graph sample and aggregate-attention (SAGE-A). SAGE-A first uses the simple linear iterative clustering (SLIC) algorithm to segment the original HSI, and then uses the multilevel graph sample and the aggregate-attention mechanism to aggregate the remote context of multihop neighborhoods. Jiang et al. [44] introduced a graph generative structure-aware transformer (GraphGST) that uses a positional encoder and stacked transformer encoders to provide local and global views on pixel representation learning, and adopts a cross-view contrastive mechanism to enable local and global views to supervise and complement each other. Although graph-based semi-supervised classification methods are simple and efficacious, the construction of graphs still does not essentially emphasize the influence of uncertainty.

Generative adversarial networks (GANs) [45], [46], [47], [48] are also attractive in SHIC. Zhan et al. [49] designed a 1-D GAN model for SHIC (HSGAN). The model uses unlabeled samples to train the 1-D GAN, after the entire model is trained, a softmax layer is used to replace the top layer of the discriminator, transforming the model into a classifier. Zhu et al. [50] constructed a semi-supervised convolutional GAN for HIC, which employs a deep 3-D CNN to generate so-called pseudo-cube samples and designs another 3-D deep residual network to discriminate the inputs. Then, the generated samples, labeled samples, and unlabeled samples are fed into the discriminator for joint training, and the trained discriminator can judge the class label and authenticity of the samples. GAN sounds promising, but its training is unstable, making it difficult to reach Nash equilibrium and prone to mode collapse.

The aforementioned semi-supervised classification approaches entail certain issues that are arduous to surmount and possess high algorithmic complexity. Inspired by bidirectional encoder representations from transformer (BERT) [51], a masked language representation model, He et al. [52] proposed MAE by analyzing the differences between visual and language masked auto-encoding. As a powerful visual self-supervised representation learning (SSRL) model, MAE was introduced into SHIC [53], [54], [55]. Mohamed et al. [56] proposed a factorized transformer architecture, called FactoFormer, which utilizes the masked spectral transformer and spatial transformer to capture the higher order spectral–spatial correlations present in HSI data. During the fine-tuning phase, FactoFormer integrates embeddings learned from factorized transformers for classification. Cao et al. [57] built a transformer-based MAE using contrastive learning (TMAC). TMAC consists of two branches, the first of which, the naive MAE, is used to reconstruct the original sample at the pixel-level. The second branch uses a momentum encoder and standard projection head to embed the unmasked sample into the feature space, then uses a feature decoder to extract the high-level features of the reconstructed sample obtained from the first branch, and finally calculates the contrast loss between the features obtained by the encoder and decoder to force the entire TMAC to learn holistic features. Kong et al. [58] developed an instructional masked auto-encoder (IMAE) for SHIC. IMAE acquires the components of global information that are highly correlated with the target pixel through computing the masked reconstruction loss and the instructional loss between the center pixel and the embedding of the instructional token. In actuality, the contrastive loss in TMAC and the instructional loss in IMAE constitute week regularization approaches. The problem resides in the fact that a weak regularizer has restricted potential for improvement in SSRL-based SHIC.

SHIC based on MAE has made remarkable progress, however, most existing methods are only methodological rather than theoretical innovations and fail to contemplate the challenges posed by the high intraclass variances and interclass similarities that often appear in HSI data. For the reasons, designing a more robust and effective SSRL model for SHIC is of utmost importance. In this article, we revisit SSRL in SHIC and propose a regularized MAE (RMAE). RMAE incorporates a self-designed induced transformer block into the fundamental framework of MAE, utilizing visible patches to acquire the embeddings of patches with broader receptive fields, namely, representations in the form of the parse tree. The acquired embeddings are employed to reconstruct the corresponding patches, and subsequently, an induced reconstruction loss is calculated. In particular, the induced reconstruction loss is also referred to as the regularization loss. More generally, RMAE builds a more challenging self-supervision task, that is, learning high-level hierarchical features from a limited number of low-level features. In addition, by rethinking the masking operations, we have devised a simple masked CNN (MCNN)

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to profoundly disclose the principles underlying MAE, and to substantiate the efficacy and rationality of the regularization approach employed in RMAE. Finally, we introduce two metrics: the mean intraclass distance and the mean interclass distance. With these metrics, we present two criteria for more effectively evaluating the performance (the linear tunability and top-layer unfrozen tunability) of an SSRL model.

The key contributions of this work are summarized as follows.

1) RMAE is proposed to further improve the performance of SHIC based on SSRL. It is specifically designed to address the classification challenges arising from the high intraclass variances and interclass similarities commonly observed in HSI data.
2) By employing a self-designed induced transformer block, RMAE constructs a more intricate self-supervisory task that acquires hierarchical features, thereby enhancing the separability of the learned representations. Compared with MAE, RMAE has stronger SSRL capabilities while maintaining low complexity.
3) By treating the masking operation as a special pooling, MCNN is developed to deeply reveal the fundamental principles behind MAE and help demonstrate the effectiveness and rationality of RMAE. Our illuminating perspective offers a novel understanding of SSRL.
4) We give two metrics: the mean intraclass distance, and the mean interclass distance. Based on the metrics, we present two criteria to formally evaluate the performance of an SSRL model.

The rest of this article is organized as follows. Section II describes the work related to SSRL in visual classification. Section III first introduces the tunability metrics based on mean intraclass and interclass distance, and then elicits the proposed RMAE by revealing the fundamental principles behind MAE deeply. The experimental results and analysis are presented in Section IV. Section V discusses the complexity and sensitivity of RMAE. Finally, Section VI summarizes this article and gives the possible research direction.

II. Related Work

SSRL (also known as unsupervised representation learning) aims to establish models for automatically discovering patterns in data and explicitly revealing these patterns through representations [59], [60]. In this section, we briefly review the work related to SSRL to lay the groundwork for the proposed RMAE.

A. Contrastive Self-Supervised Learning

Contrastive self-supervised learning creates sample pairs by random transformations (such as pruning, resizing, and flipping), then trains the model to learn a representation that preserves the commonality between the sample pairs while discarding the instance-specific details. Ji et al. [61] proposed invariant information clustering (IIC), which trains a simple network by calculating the maximum mutual information between the class assignments of each pair of samples. The highlight of IIC is that it outputs semantic labels directly rather than high-dimensional representations that require external processing for semantic clustering. Zhao et al. [62] introduced contrastive self-supervised learning into SHIC, trained a ResNet18-based Siamese network by calculating the maximum similarity (cosine similarity) between sample pairs, and finally used limited labeled samples for supervised optimization. Contrastive self-supervised learning has been widely investigated because of its simplicity and effectiveness, but the shortcoming is also obvious, that is, it needs a large number of randomly transformed samples.

B. Stacked Capsule Auto-Encoders (SCAEs)

Kosiorek et al. [63] designed the third-generation capsule network, namely, the SCAE. SCAE consists of a part capsule auto-encoder (PCAE) and an object capsule auto-encoder (OCAE). Among them, PCAE first learns the pose, presence, and special features of each part capsule, then uses fixed templates (trained by the network) to model the image as a spatial Gaussian mixture, and finally uses the Gaussian mixture to calculate image likelihood. OCAE first uses the part encodings (only special features have gradients) and gradient-detached templates as inputs to the set transformer [64] to learn the encoding features of object capsules. Then, the encoding features of object capsules are used to learn the dynamic object-part-relationships (OP), object-viewer-relationships (OV), presence probabilities, probabilities of objects-to-parts votes, and standard deviations of votes. Finally, the learned OV is multiplied by a static OP plus the learned dynamic OP to obtain the objects-to-parts votes, and the part likelihood is calculated based on the votes, presence probabilities of the object capsules, probabilities of votes, and standard deviations of votes. Shortly after the introduction of SCAE, it was also used in the study of unsupervised face part discovery [65]. In addition, Pan et al. [66] applied SCAE to unsupervised HIC. SCAE has achieved good representation on the MNIST dataset, but it is difficult to model real data effectively due to the use of fixed templates that are not sufficiently expressive. Most importantly, SCAE does not have the ability to perform top-layer unfrozen fine-tuning, and the training becomes difficult due to the use of logarithmic operations (image likelihood and part likelihood).

C. Masked Auto-Encoders (MAEs)

By analyzing the differences between visual and language-masked auto-encoding, He et al. [52] pointed out that images, as natural signals, have high spatial redundancy, e.g., with little high-level understanding of parts, objects, and scenes, it is possible to recover a missing patch from adjacent patches quite well. Furthermore, to enable the learned representations to infer complex, holistic reconstructions, MAE's decoder uses a lightweight ViT. With a high masking ratio (e.g., 75%) and asymmetric encoder–decoder structure, MAE not only greatly reduces computation, but also achieves excellent top-layer unfrozen fine-tuning performance. The statement can be made that MAE is impeccable, and MAE-based semi-supervised classification methods are gradually gaining prominence in HIC [56], [57], [58]. How to further improve MAE's learning capability will be an ongoing research focus in SHIC.

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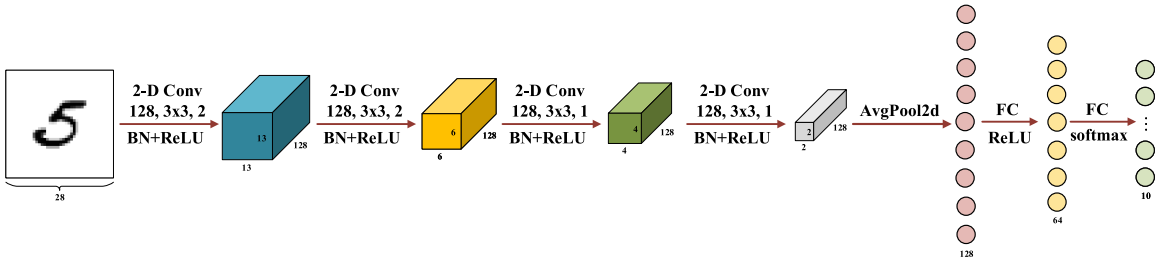


Fig. 1. Simple CNN for MNIST classification, where BN represents batch normalization and FC represents the fully connected layer.

III. Methodology

A general goal of SSRL is to build a model that transforms the raw data into a new space so that similar things are placed close to each other while dissimilar things are separated from each other, and the performance of an SSRL model is mainly reflected in its linear tunability and top-layer unfrozen tunability. However, there is no clear common principle as a benchmark to evaluate the performance of an SSRL model.

In this section, we will first cover the mean intraclass and interclass distance, and give two criteria that are used to evaluate the linear tunability and top-layer unfrozen tunability of an SSRL model. Next, by considering the masking operation as a special pooling, we construct a simple MCNN, and show that masking (pooling) operations help discover patterns, thereby revealing the principle behind MAE. Finally, based on the above proof, we introduce the proposed RMAE in detail.

A. Mean Intraclass and Interclass Distance

We usually refer to all the layers before the softmax layer of a CNN-based classifier as the feature extraction module. In fully supervised HIC tasks, the main goal is to design an excellent feature extraction module to extract the most discriminative features, therefore, the discriminative power of features requires a metric to measure. The mean intraclass and interclass distance we define, although derived from supervised classification, are also applicable to SSRL.

Let $\boldsymbol{r} \in \mathcal{R}^{n\times d}$ represent the representations that a model learns from the samples (where n represents the number of samples and d represents the embedding dimension), y represent the labels, and c represent the number of classes. The central representation of the kth class samples can be expressed as

$$\bar{\boldsymbol{r}}_k = \frac{\sum_i \boldsymbol{r}_k^i}{n_k} \tag{1}$$

where $\boldsymbol{r}_k^i$ represents the representation of the ith sample of the kth class, and n_k represents the number of the kth class samples. Then, the mean intraclass distance $\bar{d}_{\text{intra}}$ can be formulated as

$$\bar{d}_{\text{intra}} = \frac{1}{c}\sum_{k=1}^{c}\left(\frac{\sum_i \|\boldsymbol{r}_k^i - \bar{\boldsymbol{r}}_k\|_{\text{F}}}{n_k}\right) \tag{2}$$

and the interclass distance $\bar{d}_{\text{inter}}$ can be calculated as

$$\bar{d}_{\text{inter}} = \frac{2\sum_{j\neq k}^{c} \|\bar{\boldsymbol{r}}_j - \bar{\boldsymbol{r}}_k\|_{\text{F}}}{c(c-1)} \tag{3}$$

Algorithm 1 Mean Intraclass and Interclass Distance

Input: representations: $\boldsymbol{r}$; labels: y; the number of classes: c.

Output: mean intraclass distance: $\bar{d}_{intra}$; mean interclass distance: $\bar{d}_{\text{inter}}$.

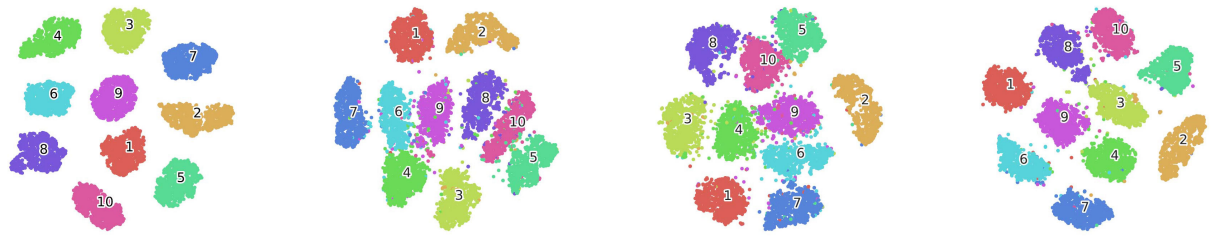
```
def mean_distance(r, y, c):
    codings_per_class = []
    for i in range(c):
        codings_per_class.append(r[y == i,:])
    centers = [torch.mean(codings_per_class[i], 0) for i in range(c)]
    d_intra = torch.tensor(0.)
    for i in range(c):
        per_d_intra = torch.tensor(0.)
        codings = codings_per_class[i]
        center = centers[i]
        num = codings.size(0)
        for j in range(num):
            per_d_intra += torch.dist(codings[j], center, p=2)
        per_d_intra = per_d_intra / num
        d_intra += per_d_intra
    d_intra_mean = d_intra / c
    d_inter = torch.tensor(0.)
    for i in range(c - 1):
        for j in range(i + 1, c):
            d_inter += torch.dist(centers[j], centers[i], p=2)
    d_inter_mean = d_inter / (c * (c - 1) / 2)
    return d_intra_mean, d_inter_mean
```

where $\|\cdot\|_{\text{F}}$ means the Frobenius norm. To further understand these two metrics, we illustrate the calculation process in Algorithm 1 in the form of PyTorch pseudo-code.

Let's take the MNIST dataset as an example to comprehend the significance of the proposed metrics. Suppose that, 10 000 samples are randomly selected from the MNIST dataset and used to train a simple CNN as shown in Fig. 1. After 100 training epochs, we collected the output of the feature extraction module (the output before the softmax layer). To better observe the separability, we present the 2-D t-distributed stochastic neighbor embedding (t-SNE) embeddings for all learned features (representations) in Fig. 2(a). At this point, by calculation, we can get $\bar{d}_{\text{intra}} = 6.0605$ and $\bar{d}_{\text{inter}} = 25.6998$. It is evident that for fully supervised models, the features learned have relatively small $\bar{d}_{\text{intra}}$ and large $\bar{d}_{\text{inter}}$. By analogy, for an SSRL model, we provide the following two criteria to evaluate its performance (linear tunability and top-layer unfrozen tunability).

Criterion 1: If an SSRL model has relatively low $\bar{d}_{\text{intra}}$ and high $\bar{d}_{\text{inter}}$, then it must have good tunability.

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(a) (b) (c) (d)

Fig. 2. Visualization of the features (representations) learned by various models. (a) Fully supervised CNN. (b) MCNN. (c) MAE. (d) RMAE.

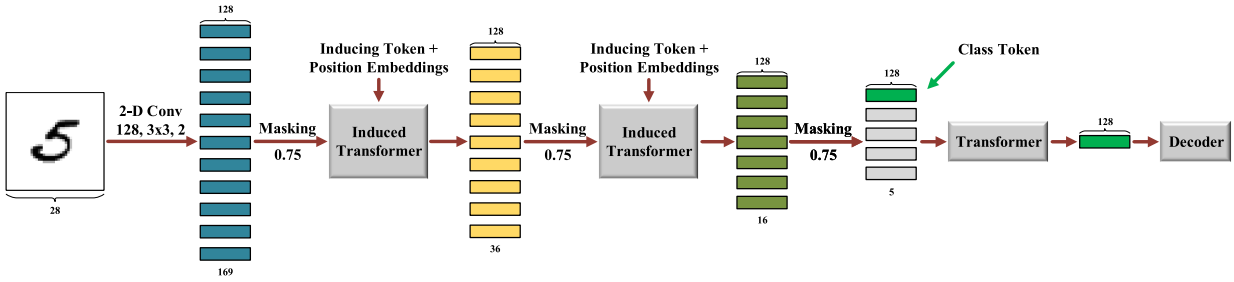


Fig. 3. MCNN, where the decoder consists of a fully connected layer and several transposed convolution layers for reconstructing the input.

Criterion 2: If an SSRL model has relatively high $\bar{d}_{\text{intra}}$ and low $\bar{d}_{\text{inter}}$, then it must be poorly tunable or even un-tunable.

Next, we introduce an interesting thing that pooling operations help discover patterns. By deeply revealing the principles behind MAE, we provide the basic idea for constructing RMAE.

B. Special Pooling

Pooling operations in CNN, including average pooling and maximum pooling, are generally considered as a way to reduce the spatial size of feature maps. However, their greatest strength lies in discarding specific features of samples to learn a universal pattern. In addition, the masking operation with a 75% masking ratio adopted by MAE can be regarded as a special pooling operation with a receptive field of 2×2 and a step of 2. To demonstrate that the masking (pooling) operations help discover patterns, we designed a simple auto-encoder that mimics the feature extraction behavior of CNN and evaluates its SSRL performance. For ease of description, we refer to this simple auto-encoder as MCNN, and the basic diagram is presented in Fig. 3.

As is shown in Fig. 3, similar to the simple CNN illustrated in Fig. 1, MCNN first uses a 2-D convolution to get the underlying embeddings, whose size is of $(N, 128, 13, 13)$, where N is the number of samples trained in batch. Then, the obtained feature maps are unfolded in the spatial dimension and fixed position embeddings are added. At this point, the embeddings can be represented as $\boldsymbol{x} \in \mathcal{R}^{N\times 169\times 128}$. Before learning the features of the next layer, a masking operation with a 75% masking ratio is adopted, and the resulting features can be expressed as $\boldsymbol{x}' \in \mathcal{R}^{N\times 42\times 128}$. The next step involves employing an induced transformer to mimic the behavior of the simple CNN (see Fig. 1), enabling us to acquire all features within a spatial size of 6×6. Besides that, to provide position information for all features, we also use fixed position embeddings in this layer. The structure of the designed induced transformer is given in Fig. 4.

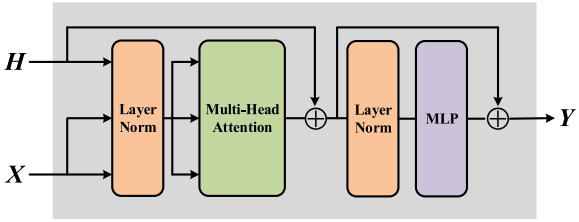


Fig. 4. Induced transformer, where MLP stands for multilayer perceptron.

Assuming the embedding dimension D at the next layer is 128, the inducing token can be expressed as $\boldsymbol{I} \in \mathcal{R}^{1\times 1\times 128}$ (it is initialized via a plus distribution having a mean of 0 and a standard deviation of 0.02), and the position embeddings can be denoted as $\boldsymbol{p} \in \mathcal{R}^{1\times L\times 128}$, where $L = 36$ denotes the number of embeddings after expansion. At this point, the input $\boldsymbol{X}$ in Fig. 4 is $\boldsymbol{x}'$, and the input $\boldsymbol{H}$ can be represented as

$$\boldsymbol{H} = \boldsymbol{I}.\text{repeat}(N, L, 1) + \boldsymbol{p}. \tag{4}$$

Specifically, the output embeddings $\boldsymbol{Y} \in \mathcal{R}^{N\times L\times D}$ is obtained through aggregation learning. First, $\boldsymbol{X}$ and $\boldsymbol{H}$ are fed into the layer normalization layer

$$\boldsymbol{H} = \text{LayerNorm}(\boldsymbol{H}) \tag{5}$$

$$\boldsymbol{X} = \text{LayerNorm}(\boldsymbol{X}). \tag{6}$$

Note that, if layer normalization has been calculated before $\boldsymbol{X}$ is fed in, its two branches will be directly fed into the multihead attention (MHA) module. In MHA, the queries $\boldsymbol{Q} \in \mathcal{R}^{N\times h\times L\times d}$ are obtained by h different projections (multihead projections) on $\boldsymbol{H}$

$$\boldsymbol{Q} = \text{Project}(\boldsymbol{H}) \tag{7}$$

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and both the keys $\boldsymbol{K} \in \mathcal{R}^{N\times h\times 42\times d}$ and values $\boldsymbol{V} \in \mathcal{R}^{N\times h\times 42\times d}$ are obtained by h different projections on $\boldsymbol{X}$

$$\boldsymbol{K} = \text{Project}(\boldsymbol{X}) \tag{8}$$

$$\boldsymbol{V} = \text{Project}(\boldsymbol{X}) \tag{9}$$

where d (the default value is 16) represents the dimension of each projection. Next, an MHA result is obtained

$$\text{Att}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\mathrm{T}}}{\sqrt{d}}\right)\boldsymbol{V}. \tag{10}$$

Finally, the obtained result $\text{Att}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) \in \mathcal{R}^{N\times h\times L\times d}$ is concatenated and a linear layer is used to adjust the dimension to match the embedding dimension D of the input $\boldsymbol{H}$. Now, let $\boldsymbol{H}'$ represent the output of MHA, then the final output $\boldsymbol{Y}$ can be expressed as

$$\boldsymbol{Y} = \left(\boldsymbol{H} + \boldsymbol{H}'\right) + \text{MLP}\left(\text{LayerNorm}\left(\boldsymbol{H} + \boldsymbol{H}'\right)\right). \tag{11}$$

With the output of the second layer, we get the output of the third layer in the same way, which corresponds to all features within a spatial size of 4×4. After the last masking operation, there are four embeddings left, at which point we add a class token to the header and feed them into the transformer (four blocks and each block uses eight-head attention) to learn the optimal representation for reconstruction.

Also taking the 10 000 MNIST digits as an example, MCNN uses the Adam optimizer with an initial learning rate (LR) of 0.001, and after being trained 200 epochs, the representations it has learned are shown in Fig. 2(b). In addition, $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ obtained by it are 5.2987 and 5.2004, respectively. From the results, $\bar{d}_{\text{intra}}$ obtained by MCNN is larger than $\bar{d}_{\text{inter}}$, which indicates that its performance (tunability) is not good. For comparison, we applied MAE to the above experiment, and the detailed configuration is that, the patch size is 2×2, the encoder has an embedding dimension of 128 and uses 12 ViT blocks (each block uses eight-head attention), and the decoder has an embedding dimension of 32 and uses eight ViT blocks. After being trained 200 epochs, the representations obtained by MAE are shown in Fig. 2(c), and $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ obtained by it are 3.5275 and 4.5076, respectively. The results show that MAE not only can learn separable representations but also has good tunability. Although MCNN is not as good (tunable) as MAE, one thing worth acknowledging is that, with masking operations, MCNN can also perform effective SSRL. Since both masking operations and pooling operations discard many detailed features to some extent, we say that pooling operations help discover patterns. However, it is notable that the masking operations randomly discard a considerable number of features in the spatial dimension of the feature maps, while the pooling operations uniformly preserve the most prominent or average features within each local receptive field along the channel dimension. In a sense, masking operations are more random and discard more features compared with pooling operations. Apparently, the masking operations merit more exploration in pattern recognition.

What MCNN and MAE share in common is that they both acquire the patterns by discarding a large number of input-dependent features, and the biggest difference between them is that MCNN learns the representation by reconstructing the entire input, while MAE aggregates the desired representation through reconstructing a large number of missing patches from a small number of visible patches. From the reconstruction perspective, MCNN allows for detail deviation from the target, while MAE is to recover as much detail as possible. More strictly speaking, MAE does not allow for detail deviations at the positions of missing patches. Therefore, the learning task of MAE is much more difficult than that of MCNN, which is also the reason why MAE performs better than MCNN.

The most crucial question is, *why does MAE only add losses on missing patches*? In other words, *what would happen if MAE also adds losses to visible patches*? Since neural networks are always good at one-to-one encoding-decoding tasks, if losses are also added to the visible patches, the reconstructed image will always be exactly recovered at the visible paths, which will affect the semantic richness (semantic rank) of the learned representations of missing patches and class token (i.e., the final representation). From the metric perspective, when we add losses to the visible patches, $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ obtained by MAE are 4.2632 and 4.9458, respectively. Compared with the above, it can be found that although $\bar{d}_{\text{inter}}$ increases by 0.4382, $\bar{d}_{\text{intra}}$ increases by 0.7357, which will degrade the tunability of MAE.

After fully understanding the principles behind MAE, to further improve its performance, we ask MAE to complete a more challenging task, namely, induced reconstruction. Now, turn attention to RMAE.

C. Regularized MAE

MAE imposes a challenging task for SSRL, recovering 75% missing patches using 25% visible patches. To further promote MAE, we create a more difficult task on the basis of MAE, that is, using a few visible patches to reconstruct patches with larger receptive fields. The losses on patches with larger receptive fields are called induced reconstruction losses, or regularization losses. The principle of RMAE is simple, no detail bias is allowed either at the position of missing patches or in larger receptive fields. Moreover, its implementation is also simple, using an induced transformer to learn representations of patches with larger receptive fields (representations in the form of a parse tree) to build the regularization loss. For ease of understanding, we present the overview of RMAE in Fig. 5.

As shown in Fig. 5, RMAE is composed of two branches behind the ViT encoder. The first branch is used to reconstruct missing patches using the ViT decoder, and the second branch is used to reconstruct patches with larger receptive fields by induced transformer.

Let $\boldsymbol{x} \in \mathcal{R}^{N\times B\times H\times W}$ (here, B denotes the number of spectral bands retained after PCA processing) represent the input HSI samples in one batch, the smaller receptive field (patch size) is k_1, and the larger receptive field is k_2. Through partitioning, we know that there are $L_1 = HW/k_1^2$ smaller patches and $L_2 = HW/k_2^2$ larger patches. First, to get the embeddings of all the smaller patches, $\boldsymbol{x}$ is fed into a 2-D convolution layer

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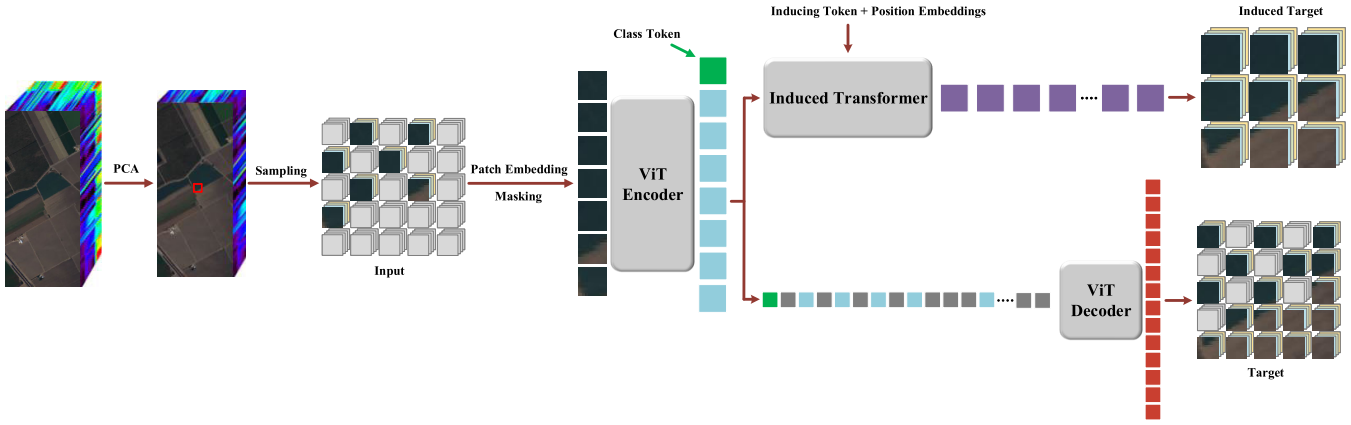


Fig. 5. Overview of the proposed RMAE framework, where PCA represents principal component analysis.

and then reshaped

$$\boldsymbol{x} = \text{Reshape}(\text{Conv2d}(B, 128, k_1, k_1)(\boldsymbol{x})). \tag{12}$$

At this point, the shape of the embedded $\boldsymbol{x}$ is $(N, L_1, 128)$. After the masking operation with a ratio of 75%, the number of retained embeddings in each sample is $L_1' = L_1//4$, and we represent the embeddings at this time as $\boldsymbol{x} \in \mathcal{R}^{N\times L_1'\times 128}$. Next, $\boldsymbol{x}$ is fed into the ViT encoder to aggregate the desired representation, i.e., the class token

$$\boldsymbol{x}' = \text{LayerNorm}(\text{Encoder}(\boldsymbol{x})). \tag{13}$$

The resulting $\boldsymbol{x}'$ has the shape of $(N, 1+L_1', 128)$, which is the input of the two downstream branches. Since the first branch represents the decoder of MAE, we will mainly focus on the second branch.

Assuming the larger patches have an embedding dimension of 128, then the inducing token can be expressed as $\boldsymbol{I} \in \mathcal{R}^{1\times 1\times 128}$, and the position embeddings can be denoted as $\boldsymbol{p} \in \mathcal{R}^{1\times L_2\times 128}$. This moment, the input $\boldsymbol{X}$ of induced transformer (see Fig. 4) is $\boldsymbol{x}'$, and the input $\boldsymbol{H}$ can be expressed as

$$\boldsymbol{H} = \boldsymbol{I}.\text{repeat}(N, L_2, 1) + \boldsymbol{p}. \tag{14}$$

It is important to note that both branches of $\boldsymbol{X}$ will be fed directly into the MHA module because it has been layer normalized before being fed into the induced transformer. Please refer to (7)–(11) for the following calculation process, which will not be repeated here. The obtained $\boldsymbol{Y} \in \mathcal{R}^{N\times L_2\times 128}$ represents the representations of patches with larger receptive fields. At last, a linear layer is used to reconstruct patches with larger receptive fields.

The loss function of RMAE is defined as

$$L = L_{\text{mr}} + \lambda L_{\text{ir}} \tag{15}$$

where L_{mr} represents the masked reconstruction loss of missing patches, L_{ir} represents the induced reconstruction loss of patches with larger receptive fields, i.e., regularization loss, and λ is the regularization coefficient.

Finally, for comparison, we apply RMAE to the 10 000 MNIST digits, and the detailed configuration is that, the smaller patch size (receptive field) k_1 is 2×2, the larger patch size k_2 is 4×4, the dimension of the inducing token is set to 128, the regularization coefficient λ is set to 0.1, and the rest of the configuration is the same as MAE. After being trained 200 epochs, the representations learned by RMAE are shown in Fig. 2(d), with $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ of 6.1486 and 8.2853, respectively. The mean intraclass and interclass distance can only qualitatively evaluate the tunability of an SSRL model. We used 100 randomly selected samples as labeled tuning samples, 59 900 samples as test samples, and 10 000 samples as validation sets to quantitatively evaluate the fine-tuning performance of various SSRL models. Table I presents the classification performance of fully supervised CNN (10 000 training samples, 50 000 test samples, and 10 000 validation samples) and the top-layer unfrozen fine-tuning[1] performance of different SSRL models, including MAE with losses on visible patches (represented as MAE w/ full loss).

TABLE I
CLASSIFICATION PERFORMANCE OF FULLY SUPERVISED CNN AND TOP-LAYER UNFROZEN FINE-TUNING PERFORMANCE OF DIFFERENT SSRL MODELS ON THE MNIST DATASET

Method	$\bar{d}_{intra}$	$\bar{d}_{inter}$	valid_acc	test_acc
Fully supervised CNN	6.0605	25.6998	99.04	98.84
MCNN	5.2987	5.2004	89.47	88.35
MAE	3.5275	4.5076	93.70	92.82
MAE w/ full loss	4.2632	4.9458	86.95	85.48
RMAE	6.1486	8.2853	96.45	95.93

When we use the randomly selected 100 tuning samples as the training samples for the fully supervised CNN, the validation accuracy and test accuracy obtained are 83.61% and 82.20%, respectively. This result fully demonstrates the superiority of various SSRL models, that is, they can autonomously learn patterns from a large number of unlabeled samples and achieve satisfactory classification results with only a small number of tuning samples. In addition, from Table I, we can see that RMAE has achieved much better results than MAE, which fully verifies the robustness and progressiveness of RMAE.

[1]As MAE did, we only tuned the last transformer block and added two linear layers (with dimensions of 64 and the number of categories) to the top of the encoder to complete the classification task.

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After all, MNIST is an easily learned dataset. However, due to the high intraclass variances and interclass similarities in HSI data, SSRL on HSI samples is quite challenging. Next, we will verify the effectiveness and superiority of RMAE on four publicly available HSI datasets.

IV. Experimental Results and Analysis

We performed rigorous comparative experiments on four HSI datasets to verify the effectiveness and superiority of RMAE. The compared models include five supervised classification models: A2S2K-RN [13], DMCN [14], MASSFormer [17], CapsViT [19], SSFTT [21], and three semi-supervised classification models: MCNN (designed by us), MAE [52], IMAE [58]. The HSI datasets we used include: Kennedy Space Center (KSC), Houston University 2013 (HU2013), HanChuan, and Salinas. Normally, we use overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) to evaluate the classification performance. In addition, when evaluating MCNN, MAE, IMAE, and RMAE, we also incorporate two newly proposed metrics: mean intraclass distance $\bar{d}_{\text{intra}}$ and mean interclass distance $\bar{d}_{\text{inter}}$. All the experiments were performed on a system with an NVIDIA GeForce RTX 3090 GPU accelerator (with 24-GB GPU memory).

A. Experimental Setup

First, the spatial size of samples used in all models is 15 × 15, as too large a spatial size leads to a large number of redundant calculations, while too small a spatial size limits the spatial information contained in samples. Second, PCA was used to reduce the dimensionality of all the datasets to 15 bands. For A2S2K-RN, MASSFormer, and SSFTT, due to the presence of 3-D convolution kernels, 30 bands were retained after dimensionality reduction. It is notable that PCA is a straightforward and efficacious dimensionality reduction approach, which projects the original data in the direction that retains the greatest difference, minimizing the loss of information as much as possible. PCA dimensionality reduction is indispensable as it enables various models to circumvent the potential issue of the curse of dimensionality, while significantly enhancing their training and testing speed. Next, we mainly introduce some hyper-parameter configurations of MCNN, MAE, IMAE, and RMAE, the hyper-parameter adjustment of other models will not be elaborated further.

1) The convolution step of the first 2-D convolution layer in MCNN (see Fig. 3) changes to 1, and the other parameter settings remain unchanged.
2) The patch size adopted by MAE is 3 × 3. Besides that, the encoder in MAE has an embedding dimension of 128 and uses 12 ViT blocks (each block uses eight-head attention), and the decoder has an embedding dimension of 64 and uses eight ViT blocks.
3) The patches processed by IMAE remains at 3 × 3. The encoder in IMAE has an embedding dimension of 256 and uses four ViT blocks (each block uses eight-head attention), and the decoder has an embedding dimension of 64 and uses two ViT blocks. The regularization coefficient α in the loss function (see [58]) was consistently fixed at 1.
4) The smaller patch (receptive field) size used by RMAE is 3 × 3 and the larger one is 5 × 5. In addition, the encoder and decoder in RMAE are configured exactly as MAE, and the embedding dimension of the inducing token is 128. Most importantly, the regularization coefficient λ was set to 1, and we will discuss the effect of λ on RMAE in Section V.

TABLE II
Some Other Settings for MCNN, MAE, IMAE, and RMAE

Model	Initial LR	LR Scheduling	Optimizer	Pre-Training Epoch
MCNN	1e-3	$\eta \times 0.99^{\text{epoch}}$	Adam	400
MAE	1e-3	$\eta \times 0.99^{\text{epoch}}$	Adam	400
IMAE	1e-3	$\eta \times 0.99^{\text{epoch}}$	Adam	400
RMAE	1e-3	$\eta \times 0.99^{\text{epoch}}$	Adam	400

For clarity, Table II presents the initial LR η, LR scheduling, optimizer, and pretraining epoch settings for MCNN, MAE, IMAE, and RMAE. As for the pretraining batch size, it was uniformly set to 128.

Finally, for all datasets, we randomly selected ten (labeled) samples from each category as supervised (tuning) samples, and the remaining samples were used as test samples. In addition, to obtain the optimal parameters of each model, we randomly selected 100 samples from each category from the test set to build the validation set. All supervised models were trained 100 epochs with a batch size of 32, and the tuning epochs for all semi-supervised models were set to 100. Another detail that needs to be mentioned is that for a dataset such as the KSC dataset, when the last batch in one epoch contains few samples, the training of the supervised models will become unstable, in which case the batch size or training epoch can be modified. As the selection of labeled (tuning) samples will inevitably have a great impact on the performance of the model, to comprehensively evaluate the performance of each model on each dataset, we repeated the experiment ten times[2] for each model to report the AA and standard deviation.

B. Experiment 1: KSC Dataset

The KSC dataset was captured in 1996 by Airborne Visible Infrared Imaging Spectrometer (AVIRIS). A total of 224 spectral bands were collected, with wavelengths ranging from 400 to 2500 nm. After removing water absorption and low signal-to-noise ratio (SNR) bands, 176 bands were reserved for analysis. Moreover, the KSC dataset has a spatial resolution of 18 m and a spatial dimension of 512 × 614 pixels, and contains a total of 5211 samples in 13 categories. Fig. 6(a) exhibits the false-color composite image, and Fig. 6(b) shows the ground truth map.

Due to the low labeled pixels proportion of the KSC dataset, all samples were used as pretraining samples to learn the representations for compared semi-supervised models. We give the classification results of all models in Table III.

[2]Note that, for semi-supervised models (MCNN, MAE, IMAE, and RMAE), the ten repeated experiments mentioned here do not include their pretraining, only their tuning experiments.

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TABLE III
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE KSC DATASET

Class	Supervised Classification Models					Semi-Supervised Classification Models			
	A2S2K-RN	DMCN	MASSFormer	CapsViT	SSFTT	MCNN	MAE	IMAE	RMAE
C1	**99.95±0.09**	98.56±1.61	99.04±0.55	98.43±1.40	99.56±0.30	92.58±3.74	96.55±3.20	97.56±2.29	99.06±1.26
C2	**97.64±2.65**	85.84±9.26	89.83±7.82	88.20±7.94	89.83±8.08	75.79±5.17	90.17±10.93	92.75±9.31	94.72±5.85
C3	**99.27±1.18**	94.67±3.42	95.16±3.37	90.45±8.68	96.22±3.29	79.19±10.91	93.86±6.41	96.67±2.18	96.38±4.40
C4	**87.85±5.76**	75.99±6.07	85.70±7.29	46.78±7.76	72.60±10.29	58.68±12.56	81.57±9.86	87.64±9.53	84.75±13.19
C5	95.17±5.78	79.4±4.04	85.16±11.31	78.21±5.17	84.37±8.40	76.95±10.79	93.84±8.03	87.22±7.10	**95.37±3.44**
C6	**97.67±2.55**	91.96±5.53	91.46±5.56	81.64±12.78	91.19±10.44	85.48±7.40	94.15±4.90	91.92±9.47	97.44±3.22
C7	**100.00±0.00**	99.68±0.95	99.68±0.95	99.26±2.21	96.31±5.60	98.10±3.70	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C8	**99.93±0.21**	74.11±8.73	96.96±3.15	84.84±8.70	99.38±0.80	66.10±11.43	94.89±6.70	98.31±0.90	96.48±4.93
C9	97.20±5.05	92.49±7.02	95.22±6.30	96.82±5.42	94.41±6.43	91.29±7.47	97.33±3.58	96.98±4.36	**97.51±3.66**
C10	99.82±0.53	93.30±2.92	99.04±1.67	93.71±5.16	97.56±3.08	94.32±4.79	99.49±1.36	96.60±5.30	**100.00±0.00**
C11	99.66±1.03	96.62±3.60	99.44±0.61	99.36±0.83	99.07±1.20	98.63±0.85	**100.00±0.00**	99.78±0.07	**100.00±0.00**
C12	**99.01±1.56**	97.97±2.11	90.26±8.44	85.03±9.70	90.28±7.04	84.81±7.92	95.68±8.42	92.74±6.68	95.56±3.29
C13	99.89±0.33	**100.00±0.00**	99.52±1.44	99.55±0.60	99.99±0.03	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA(%)	**98.58±0.43**	92.85±1.47	95.78±1.87	91.13±1.98	95.25±1.19	87.77±1.92	96.28±1.35	96.40±1.08	97.50±1.12
AA(%)	**97.92±0.48**	90.82±1.67	94.35±2.39	87.87±2.42	93.14±1.49	84.76±2.34	95.20±1.69	95.24±1.40	96.71±1.24
κ	**0.9842±0.005**	0.9203±0.016	0.9529±0.021	0.9011±0.022	0.9470±0.013	0.8636±0.021	0.9585±0.015	0.9599±0.012	0.9721±0.013

Accuracies(%)

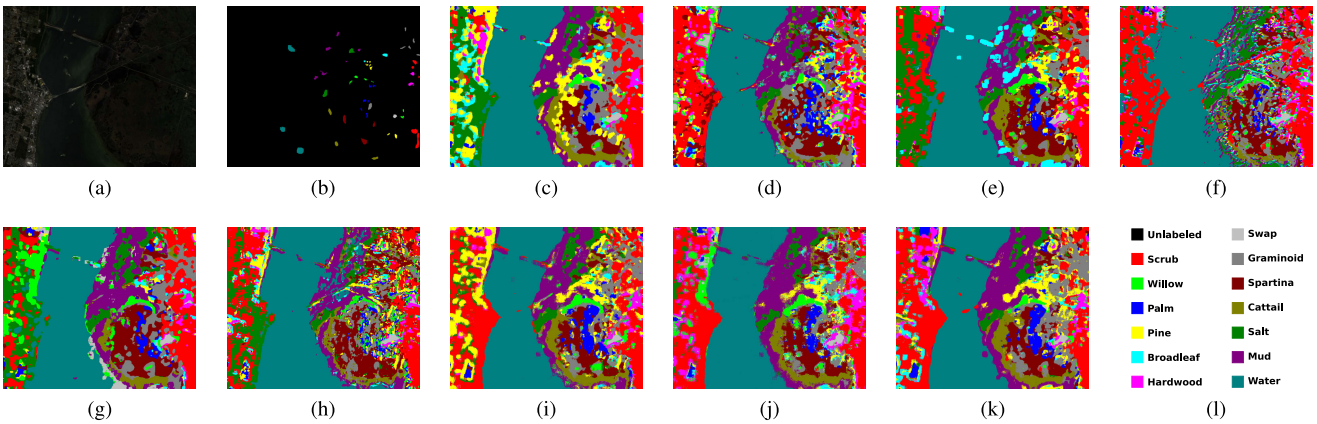


Fig. 6. Visualization of the classification maps obtained by different models on the KSC dataset. (a) Three-band false-color composite. (b) Ground truth. (c)–(k) Classification maps obtained by A2S2K-RN, DMCN, MASSFormer, CapsViT, SSFTT, MCNN, MAE, IMAE, and RMAE. (l) Legend.

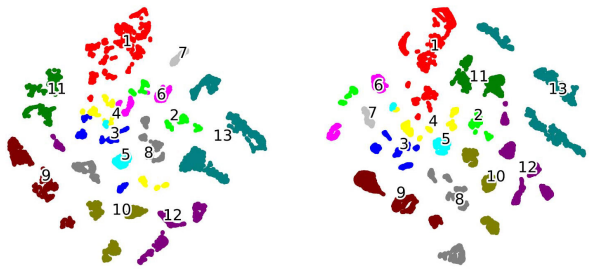
As is shown in Table III, thanks to the adaptive spectral–spatial kernel attention module which can adaptively extract spectral–spatial joint features with different receptive fields, A2S2K-RN achieved the best performance with ten labeled samples per class. Compared to other supervised models, A2S2K-RN shows clear advantages, not only in terms of performance, but also in terms of stability, which can be clearly observed from its standard deviations on OA, AA, and κ (0.43, 0.48, and 0.005). In addition, MASSFormer achieved slightly better performance than SSFTT. As a newly developed classification model in the remote sensing community, MASSFormer has demonstrated impressive strength, thanks to its use of memory tokens obtained through pooling operations in each transformer encoder to preserve memory.

By comparing MCNN, MAE, IMAE, and RMAE, we can see that the performance of MCNN that allows for reconstruction to deviate from the target does not perform well, which also indicates that a simple learning task cannot enable an SSRL model to learn good representations. That's also why we introduce induced reconstruction loss to build RMAE on the basis of MAE. RMAE creates a harder learning task, namely, having visible patches learn not only representations of surrounding missing patches, but also representations of patches with larger receptive fields. It is essential to note that we merely only employed a single-layer induced transformer to construct the induced reconstruction loss (regularization loss), which will not impose a significant computational burden on the underlying MAE. Even compared to A2S2K-RN, RMAE was still able to get the highest recall rate in six categories, which speaks volumes about its superiority. Moreover, through the comparison between IMAE and MAE, it can be discovered that IMAE performs marginally better than MAE, which suggests that the approach of IMAE in constructing the instructional loss by calculating the deviation between the instructor (the embedding of the instructional token) and the center pixel is effective. In brief, IMAE explicitly accentuates the significance of the reconstruction of the central pixel, nevertheless, what it contributes to MAE is a feeble regularization term, thus its enhancement to MAE is not conspicuous. In contrast to IMAE, RMAE can achieve strikingly superior performance. This is not only because it also implicitly incorporates the pixel-level reconstruction of the center pixel (refer to Fig. 5), but also because it establishes a more demanding self-supervised task.

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TABLE IV
MEAN INTRACLASS AND INTERCLASS DISTANCE OF THE REPRESENTATIONS LEARNED BY MCNN, MAE, IMAE, AND RMAE ON THE KSC DATASET

Method	$\bar{d}_{intra}$	$\bar{d}_{inter}$	OA(%)	AA(%)	$\bar{\kappa}$
MCNN	2.3782	3.8830	87.77	84.76	0.8636
MAE	6.3721	10.7668	96.28	95.20	0.9585
IMAE	7.2800	12.9402	96.40	95.24	0.9599
RMAE	4.3893	8.9364	97.50	96.71	0.9721



(a) (b)

Fig. 7. Two-dimensional t-SNE embeddings of the representations learned by MAE and RMAE on the KSC dataset. (a) MAE. (b) RMAE.

Classification accuracy can quantitatively evaluate the performance of various models, and when there are no test samples available for evaluation, classification maps can be used to qualitatively evaluate each model. We used a random seed (42) to select fixed training samples to train each model to obtain classification maps, which are shown in Fig. 6. Taking category 12 (purple) as a sample, it is obvious that the classification maps obtained by A2S2K-RN, DMCN, MAE, IMAE, and RMAE are smoother. Looking back at Table III, we can also see that they all have a high recall rate for category 12, which are 99.01%, 97.97%, 95.68%, 92.74%, and 95.56%, respectively. Moreover, from the right half, we can clearly find that A2S2K-RN and RMAE have better texture details, which is why they can achieve satisfactory classification results on the KSC dataset.

Finally, we analyze MCNN, MAE, IMAE, and RMAE in combination with mean intraclass distance $\bar{d}_{\text{intra}}$ and mean interclass distance $\bar{d}_{\text{inter}}$ to shed light on the differences in their performance. For comparison, we give their resulting $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ in Table IV. As the data given in Table IV, although MCNN achieved the minimum $\bar{d}_{\text{intra}}$, its $\bar{d}_{\text{inter}}$ is not large enough. In contrast, MAE acquired a relatively large $\bar{d}_{\text{inter}}$ and a relatively large $\bar{d}_{\text{intra}}$, while IMAE obtained the largest $\bar{d}_{\text{inter}}$, but it also obtained the largest $\bar{d}_{\text{intra}}$. RMAE appears much more perfect, as it obtained relatively small $\bar{d}_{\text{intra}}$ and relatively large $\bar{d}_{\text{inter}}$, with $\bar{d}_{\text{inter}}$ being greater than twice $\bar{d}_{\text{intra}}$. As RMAE is an enhancement of MAE, we present the 2-D t-SNE embeddings of their learned representations in Fig. 7 for a more intuitive observation of separability.

C. Experiment 2: HU2013 Dataset

The HU2013 dataset was gathered by the National Center for Airborne Laser Mapping (NCALM) over the University of Houston and neighboring urban areas, and was published as part of the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest. There are 144 spectral bands within a wavelength range of 380–1050 nm, and the images include 349 × 1905 pixels with a spatial resolution of 2.5 m. The dataset contains 15 land-cover classes, and the presence of shadow areas poses challenges to the classification task. Fig. 8(a) and (b) shows the false-color composite and ground truth.

For this dataset, all 15 029 samples were used as pretraining samples for compared semi-supervised models, and the classification results of all models are given in Table V.

There will always be a particular classifier that performs best on a particular dataset. As is shown in Table V, IMAE achieved the best classification results on the HU2013 dataset, mainly due to its highest recall rates in Residential (class 7) and Road (class 9), with sample sizes of 1268 and 1252, respectively. This also substantiates that the regularization method adopted by IMAE is effective. In addition, it is certain that, in spite of the challenging HU2013 dataset, CapsViT still achieved remarkable performance. CapsViT's ability to achieve high classification accuracy with a small number of samples also confirms its powerful feature extraction ability, that is, using capsules to represent the probabilities and pose of entity features, and learning the global correlation between all entity features through ViT. Our proposed RMAE, while not performing as well as IMAE and CapsViT on the HU2013 dataset, also achieved the third-best result among all classification models. It is worthy of note that, RMAE showed improvements of 1.25% in OA, 0.82% in AA, and 0.013 in κ compared to MAE. Even though the learning task of MCNN is relatively simple, it still attained an OA of nearly 83%. Moreover, MASSFormer is remarkable, despite its lightweight nature, it can achieve satisfactory classification results.

Similarly, we used a random seed (43) to select fixed training samples to observe the classification maps obtained by each model on the HU2013 dataset, as shown in Fig. 8. When using the same training samples, we can see that the classification maps obtained by IMAE and CapsViT are significantly better than those obtained by other models in terms of texture details. MASSFormer and SSFTT also have good visual effects, as the accuracies of MASSFormer and SSFTT are comparable to CapsViT when using the current training samples. Finally, by comparing MAE and RMAE, we can see that on the left side, RMAE's classification effect on Road (brown, class 9) is obviously better than that of MAE, and on the right side, RMAE also has a better classification effect on Healthy Grass (red, class 1).

To better understand the effectiveness of RMAE, we evaluate its tunability using the mean intraclass and interclass distance. For comparison, we give $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ obtained by MCNN, MAE, IMAE, and RMAE in Table VI. First, it can be noted that IMAE attained the maximum value in both metrics, and its performance is highly satisfactory as per the experimental results. Second, we can see that $\bar{d}_{\text{intra}}$ obtained by MCNN is larger than $\bar{d}_{\text{inter}}$, which is why its tuning performance is not good. Furthermore, by comparing MAE and RMAE, it can be observed that the difference between

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TABLE V
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE HU2013 DATASET

Class	Supervised Classification Models					Semi-Supervised Classification Models			
	A2S2K-RN	DMCN	MASSFormer	CapsViT	SSFTT	MCNN	MAE	IMAE	RMAE
C1	84.08±6.75	86.90±5.02	87.26±5.80	**91.39±4.71**	79.01±7.23	85.07±5.65	78.91±5.56	90.02±5.76	85.66±4.59
C2	83.59±5.65	89.77±4.87	88.78±5.25	94.20±4.32	88.82±5.48	88.42±5.84	87.66±6.61	**95.31±4.14**	79.63±6.86
C3	96.91±2.80	97.72±1.69	99.54±0.27	98.14±1.35	96.96±2.54	90.99±5.54	96.55±2.47	**99.90±0.09**	97.71±3.11
C4	78.61±8.03	91.16±4.35	89.87±5.52	**93.48±4.46**	90.36±7.05	77.52±7.23	79.59±6.70	91.12±2.96	78.40±8.99
C5	**99.97±0.07**	99.53±1.11	99.85±0.36	99.83±0.25	99.70±0.57	98.34±1.57	88.36±9.35	96.06±5.30	98.08±1.76
C6	95.40±5.13	89.97±5.37	91.74±5.84	92.16±5.95	93.72±6.85	90.16±5.60	94.51±4.99	**98.73±3.81**	92.06±5.44
C7	76.68±5.86	74.32±8.43	80.35±5.46	78.86±6.59	80.10±3.97	72.12±5.93	76.28±8.82	**93.79±4.27**	78.47±4.18
C8	61.32±10.92	54.26±8.49	64.22±6.09	68.40±7.70	60.21±7.41	65.96±6.82	66.05±10.08	**70.45±7.01**	67.10±8.15
C9	75.64±9.80	74.79±8.23	66.60±8.15	72.94±9.31	68.27±9.53	68.47±9.16	72.88±9.09	**81.63±5.45**	70.41±7.54
C10	85.61±10.79	85.60±9.37	81.76±12.85	86.07±7.82	86.71±11.45	92.00±5.52	93.16±7.88	81.56±10.31	**94.91±4.94**
C11	92.16±7.82	82.38±11.22	**92.53±7.84**	83.35±8.62	85.54±10.47	81.01±8.62	89.56±6.35	79.50±11.40	89.47±7.53
C12	75.85±13.15	71.13±8.11	77.61±9.62	78.04±7.86	78.28±8.54	80.83±9.96	85.21±6.89	76.99±7.04	**91.32±3.75**
C13	92.46±4.91	88.54±3.05	90.87±2.51	89.63±2.00	**94.49±2.76**	85.93±6.07	90.37±7.08	88.21±2.23	87.93±7.94
C14	**100.00±0.00**	99.98±0.07	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	97.18±3.11	99.86±0.36	99.98±0.07	**100.00±0.00**
C15	99.04±1.02	99.26±1.44	99.17±1.22	99.95±0.10	99.69±0.92	98.11±3.38	98.32±2.69	**100.00±0.00**	98.52±3.55
OA(%)	83.99±1.97	83.51±2.21	85.27±1.54	86.69±1.20	84.34±1.37	82.96±1.13	84.18±1.98	**87.72±1.13**	85.43±1.12
AA(%)	86.49±1.60	85.69±1.82	87.34±1.24	88.43±0.95	86.79±1.17	84.81±0.92	86.49±1.68	**89.55±0.92**	87.31±0.84
κ	0.8272±0.021	0.8218±0.024	0.8409±0.017	0.8562±0.013	0.8308±0.015	0.8159±0.012	0.8292±0.021	**0.8672±0.012**	0.8425±0.012

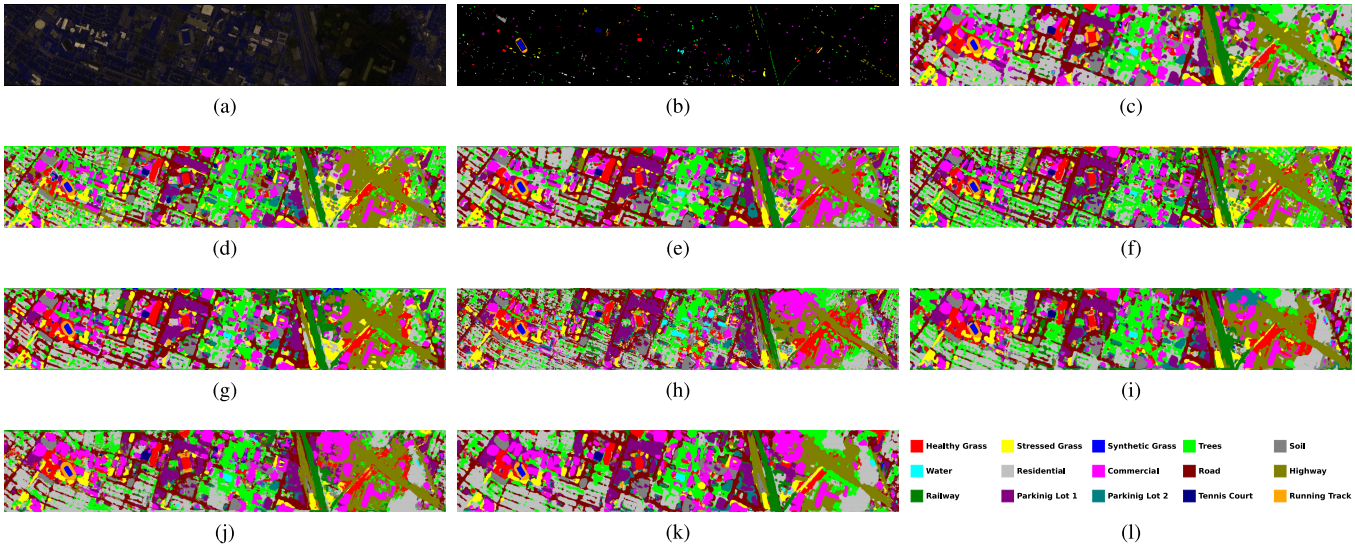


Fig. 8. Visualization of the classification maps obtained by different models on the HU2013 dataset. (a) Three-band false-color composite. (b) Ground truth. (c)–(k) Classification maps obtained by A2S2K-RN, DMCN, MASSFormer, CapsViT, SSFTT, MCNN, MAE, IMAE, and RMAE. (l) Legend.

them is not significant (using MCNN as a reference). But the problem is that for current SSRL methods, we cannot explicitly require them to reduce $\bar{d}_{\text{intra}}$ while increasing $\bar{d}_{\text{inter}}$. It is clear that RMAE achieves higher accuracy while having a smaller standard deviation compared to MAE. We speculate that for the HU2013 dataset, when two SSRL models differ little on the above two metrics, the one with a larger $\bar{d}_{\text{inter}}$ tends to have better tuning performance. Finally, to compare the identifiability of the representations learned by MAE and RMAE more intuitively, we present the 2-D t-SNE embedding maps in Fig. 9.

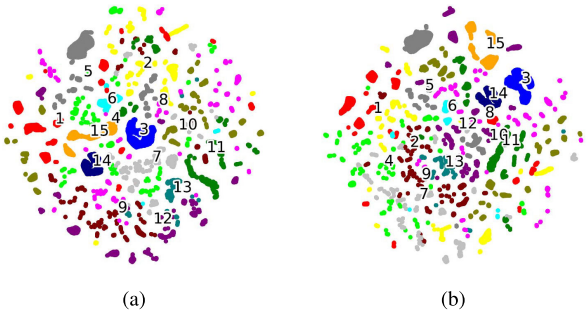


Fig. 9. Two-dimensional t-SNE embeddings of the representations learned by MAE and RMAE on the HU2013 dataset. (a) MAE. (b) RMAE.

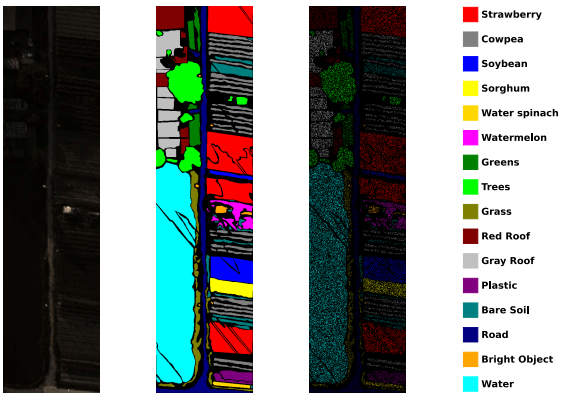
D. Experiment 3: HanChuan Dataset

The HanChuan dataset was acquired in 2016 using a Headwall Nano-Hyperspec sensor. The size of the imagery is 1217 × 303, there are 274 bands from 400 to 1000 nm, and the spatial resolution is about 0.109 m. The dataset contains 16 ground types and has a very high labeled pixels proportion. Notably, there are many shadow-covered areas in the image, which brings great challenges to the classification

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TABLE VI
MEAN INTRACLASS AND INTERCLASS DISTANCE OF THE REPRESENTATIONS LEARNED BY MCNN, MAE, IMAE, AND RMAE ON THE HU2013 DATASET

Method	$\bar{d}_{intra}$	$\bar{d}_{inter}$	$\bar{OA}(\%)$	$\bar{AA}(\%)$	$\bar{\kappa}$
MCNN	3.9853	3.7665	82.96	84.81	0.8159
MAE	7.0455	7.6820	84.18	86.49	0.8292
IMAE	9.4257	13.7294	87.72	89.55	0.8672
RMAE	7.4674	7.9501	85.43	87.31	0.8425



(a) (b) (c) (d)

Fig. 10. HanChuan dataset. (a) Three-band false-color composite. (b) Ground truth. (c) Pretraining samples. (d) Legend.

task. Fig. 10(a) and (b) shows the false-color composite and ground truth.

Due to the high proportion of labeled pixels, it is unrealistic for all samples to be used as pretraining samples. Therefore, we randomly selected one-fifth of the samples (51 506 samples in total) to be used as pretraining samples, and Fig. 10(c) shows the pretraining samples selected.

The classification results obtained by all the compared models are given in Table VII. As the data presented, RMAE achieved overwhelming performance, which fully demonstrates the charm of SSRL. We say that supervised models are dedicated to extracting distinguishable features with the presence of labels, while SSRL models are dedicated to automatically learning separable representations. When the number of labeled samples was limited, SSRL showed unmatched performance. RMAE not only obtained the highest values in three metrics, but also the highest recall rates in 11 categories. It is important to know that for the HanChuan dataset, a 1% improvement in OA means that the model can correctly classify 2575 samples.

Besides RMAE, A2S2K-RN obtained the second-best classification result, which once again proves that adaptive multiscale spectral–spatial feature extraction helps to improve accuracy. Both DMCN and CapsViT, used routing mechanisms, got similar classification results. By introducing memory tokens into transformer encoders, MASSFormer achieved satisfactory results. In fact, the memory tokens obtained through pooling operations contain input-independent pattern information, which is why the MASSFormer is so excellent. SSFTT had been lukewarm and achieved midstream classification accuracy. Despite the simplicity of the SSRL task, MCNN was able to achieve better performance than DMCN and CapsViT, which proves that efficient SSRL can also be performed using masking operations in convolutional auto-encoders. By increasing the difficulty of the learning task and emphasizing the importance of the decoder, MAE achieved a performance that was only inferior to IMAE, RMAE, and A2S2K-RN. The perfection of MAE lies not only in its ability to learn many visual concepts, i.e., semantics, but also in its excellent fine-tuning performance. For IMAE, it regards the embedding of the instructional token as an instructor and compels the model to acquire the representation that is strongly correlated with the target pixel by computing the deviation between the instructor and the central pixel. Compared with MAE, IMAE exhibits certain enhancements, however, such enhancements are not conspicuous as it merely integrates a relatively weak regularizer into MAE. The proposed RMAE did not destroy the perfect framework of the underlying MAE, but only introduced a single-layer induced transformer to learn the representations of patches with larger receptive fields through a small number of visible patches, thereby increasing the difficulty of SSRL to obtain a more holistic representation.

By using a random seed (42), we made all compared models use the same training samples and obtained the classification maps shown in Fig. 11. From the figure, we can evidently observe that the classification map obtained by CapsViT appears blurry because of misclassifying an excessive number of samples. The visual effect of the classification map derived from IMAE seems relatively favorable, yet it still misclassifies a considerable number of class 1 (strawberry, red) samples into other categories. MCNN misclassified a large number of class 2 (cowpea, dark gray) samples as class 6 (watermelon, pink) because the representations learned were not discriminative enough. Then, in terms of classification effect, A2S2K-RN and MAE are similar. Of course, the classification map generated by RMAE demonstrates the best performance, featuring not only fewer noise points but also excellent texture details.

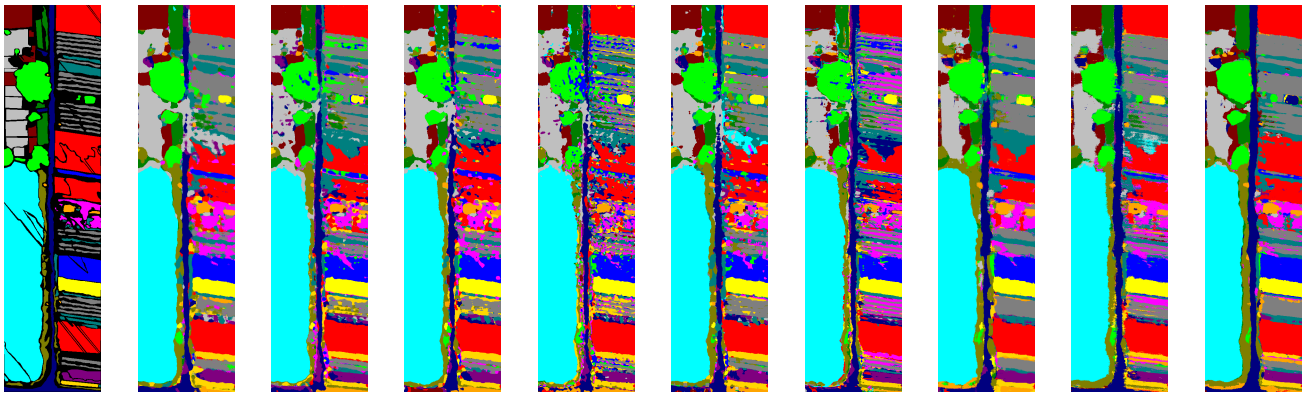
Finally, we analyze the reason why RMAE achieved such good performance by combining $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$. As the data are given in Table VIII, $\bar{d}_{\text{intra}}$ obtained by MCNN and MAE are very close, with a difference of only 0.0427. However, MAE obtained a lager $\bar{d}_{\text{inter}}$. Obviously, the representations learned by MAE are more discriminative. Although IMAE received the maximum $\bar{d}_{\text{inter}}$, it also brought the maximum $\bar{d}_{\text{intra}}$. By comparing MAE and RMAE, RMAE is 0.2961 and 0.4138 higher than MAE in the above two metrics, respectively. A large improvement in $\bar{d}_{\text{inter}}$ covers up a small deterioration in $\bar{d}_{\text{intra}}$, helping RMAE achieve excellent performance on the HanChuan dataset. For this dataset, due to the large number of pretraining samples, we will no longer present the 2-D t-SNE embeddings of the representations learned by MAE and RMAE.[3]

[3]Interested readers could reproduce the 2-D t-SNE embeddings using the source code we provide.

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TABLE VII
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE HANCHUAN DATASET

Class	Supervised Classification Models					Semi-Supervised Classification Models			
	A2S2K-RN	DMCN	MASSFormer	CapsViT	SSFTT	MCNN	MAE	IMAE	RMAE
C1	**89.24±5.98**	79.99±10.28	87.49±4.49	83.73±6.71	83.66±8.12	81.19±5.71	87.04±3.75	87.43±10.33	88.40±5.30
C2	78.22±5.51	63.21±11.21	78.40±5.03	54.18±8.96	78.69±4.43	71.76±5.35	83.03±5.93	81.89±4.70	**84.92±7.30**
C3	93.19±4.68	87.82±4.56	92.72±4.00	88.03±3.36	83.63±7.16	85.97±4.35	91.80±3.53	91.10±3.72	**93.89±3.59**
C4	98.82±0.66	97.74±1.55	97.42±1.35	97.13±1.05	97.30±1.15	97.54±1.58	96.95±2.32	97.13±1.66	**98.89±2.05**
C5	98.16±1.84	96.07±4.21	98.66±1.61	92.61±10.21	98.11±1.21	98.97±1.52	97.29±1.44	96.70±2.83	**99.16±0.92**
C6	74.12±6.87	66.70±11.38	73.32±9.20	54.59±14.20	66.93±9.40	62.17±7.27	75.37±7.96	70.79±5.43	**77.32±8.31**
C7	92.32±4.64	89.81±5.94	90.61±4.62	90.73±4.51	92.38±5.85	82.26±7.07	89.68±3.70	87.44±5.80	**93.23±3.06**
C8	72.14±7.85	63.03±7.89	75.17±6.04	64.18±5.83	64.63±7.00	74.80±4.54	76.30±5.98	74.52±5.15	**80.29±6.14**
C9	76.16±7.59	57.87±8.09	68.75±6.84	46.68±13.93	70.09±10.52	73.44±9.28	74.95±4.94	79.22±5.74	**85.25±7.43**
C10	**97.68±2.53**	93.66±5.31	96.33±2.55	93.51±4.20	93.50±5.98	93.00±1.95	94.81±3.95	95.43±1.55	95.75±2.43
C11	**92.93±5.82**	81.05±9.23	83.87±4.56	79.64±11.22	85.57±5.83	78.47±9.18	85.22±3.73	86.27±4.05	90.83±3.16
C12	89.30±7.56	79.63±12.08	83.50±6.47	79.48±12.98	84.70±7.24	73.20±7.76	91.95±5.57	88.28±7.16	**94.69±5.47**
C13	61.39±8.96	47.55±6.99	59.46±7.28	45.95±17.98	58.33±8.46	48.93±5.99	59.41±7.39	56.05±4.24	**64.93±5.38**
C14	76.42±5.42	69.95±11.86	71.38±11.40	67.63±7.64	71.60±3.88	75.40±5.73	79.85±7.30	87.62±4.37	**89.75±3.01**
C15	**91.62±5.41**	87.72±4.80	85.19±5.81	83.55±12.04	85.81±5.74	67.82±6.16	86.24±5.35	86.08±5.73	91.39±3.75
C16	97.00±1.94	90.38±6.35	97.16±1.72	96.93±1.88	96.14±2.79	**98.14±0.82**	93.50±7.25	97.08±0.89	96.58±1.21
OA(%)	87.76±1.25	79.08±3.29	86.16±1.14	79.97±1.76	84.09±2.03	83.47±0.84	86.45±2.65	87.82±1.76	**90.04±1.44**
AA(%)	86.17±0.97	78.26±2.33	83.71±1.18	76.16±1.75	81.94±1.42	78.94±1.14	85.21±1.06	85.19±1.08	**89.08±0.81**
κ	0.8576±0.014	0.7592±0.036	0.8391±0.013	0.7676±0.020	0.8151±0.023	0.8079±0.009	0.8432±0.029	0.8585±0.020	**0.8842±0.017**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 11. Visualization of the classification maps obtained by different models on the HanChuan dataset. (a) Ground truth. (b)–(j) Classification maps obtained by A2S2K-RN, DMCN, MASSFormer, CapsViT, SSFTT, MCNN, MAE, IMAE, and RMAE.

TABLE VIII
MEAN INTRACLASS AND INTERCLASS DISTANCE OF THE REPRESENTATIONS LEARNED BY MCNN, MAE, IMAE, AND RMAE ON THE HANCHUAN DATASET

Method	$\bar{d}_{intra}$	$\bar{d}_{inter}$	$\bar{OA}$(%)	$\bar{AA}$(%)	$\bar{\kappa}$
MCNN	1.3773	1.8639	83.47	78.94	0.8079
MAE	1.3346	2.6057	86.45	85.21	0.8432
IMAE	6.8380	10.1053	87.82	85.19	0.8585
RMAE	1.6307	3.0195	90.04	89.08	0.8842

E. Experiment 4: Salinas Dataset

The Salinas dataset is one of the most classical datasets in HIC research, and the experiments on this dataset are more convincing. The dataset was acquired by AVIRIS over Salinas Valley in California. The size of the imagery is 512 × 217, there are 204 bands from 400 to 2500 nm, and the spatial resolution is about 3.7 m. The scene covers 16 land-cover classes. Fig. 12(a) and (b) shows the three-band false-color composite and ground truth.

For the Salinas dataset, we randomly selected half of the samples (27 064 samples in total) as pretraining samples, and Fig. 12(c) shows the pretraining samples selected. The classification results are given in Table IX.

As we can see from Table IX, in terms of average OA and κ, A2S2K-RN performed most prominently on the Salinas dataset, MASSFormer obtained the second best result, and DMCN and SSFTT achieved similar performance. One definite point is that in terms of the average AA, the proposed RMAE is the most superior one. Compared with MAE, IMAE demonstrates certain enhancements, nevertheless, when compared with RMAE, the improvement of IMAE over MAE is restricted. It is worth noting that from the standard deviation of OA, the highest expected OA of DMCN is 96.23%, indicating that DMCN has great potential on the Salinas dataset. Moreover, it is surprising that MCNN performed slightly better than MAE. This is because the Salinas dataset

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TABLE IX
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE SALINAS DATASET

	Class	Supervised Classification Models					Semi-Supervised Classification Models			
		A2S2K-RN	DMCN	MASSFormer	CapsViT	SSFTT	MCNN	MAE	IMAE	RMAE
Accuracies(%)	C1	99.85±0.43	99.79±0.22	99.55±1.27	**99.92±0.14**	99.54±0.89	99.32±0.56	98.17±4.41	99.88±0.27	99.84±0.27
	C2	**100.00±0.00**	**100.00±0.00**	99.99±0.01	99.98±0.07	99.93±0.10	99.71±0.37	95.57±5.32	98.56±1.82	97.92±3.12
	C3	99.89±0.24	**100.00±0.00**	**100.00±0.00**	99.98±0.04	71.06±8.68	99.72±0.34	99.97±0.06	99.96±0.07	**100.00±0.00**
	C4	98.42±1.50	**99.88±0.18**	98.47±0.80	98.80±1.08	99.46±0.56	96.81±1.80	93.28±4.12	98.95±1.12	98.14±1.35
	C5	96.47±2.56	96.72±2.98	97.56±2.47	96.93±2.73	96.47±2.66	95.14±0.76	93.61±5.28	97.01±3.84	**99.44±0.38**
	C6	99.45±0.78	99.66±0.52	**99.68±0.92**	99.23±1.40	99.50±0.68	99.16±1.24	95.92±3.71	98.17±2.13	97.82±3.95
	C7	99.86±0.37	99.89±0.31	99.64±0.93	99.14±2.13	**99.95±0.05**	96.45±3.77	95.77±5.68	98.06±3.30	99.95±0.14
	C8	**87.54±4.10**	81.85±9.60	83.98±5.86	77.52±5.80	82.52±4.50	78.43±4.93	76.06±5.83	84.13±4.61	81.28±4.36
	C9	**100.00±0.00**	99.97±0.08	99.53±1.39	99.91±0.22	**100.00±0.00**	97.69±1.65	99.57±0.89	99.78±0.48	99.64±1.03
	C10	97.39±1.35	96.68±2.47	95.34±2.71	95.09±1.71	95.00±2.88	**97.39±1.07**	95.26±3.39	95.84±2.74	97.36±3.78
	C11	99.95±0.11	**100.00±0.00**	99.99±0.03	99.85±0.32	99.91±0.22	99.95±0.08	99.22±0.94	99.88±0.31	99.97±0.06
	C12	98.06±1.69	**99.91±0.09**	98.86±1.53	99.38±1.36	99.82±0.26	99.14±0.88	98.66±1.33	98.96±1.27	99.86±0.15
	C13	98.34±1.95	99.43±1.37	98.91±2.11	**99.59±0.67**	99.11±0.83	98.76±0.98	99.46±1.08	97.46±3.49	99.39±1.44
	C14	95.84±6.44	**99.75±0.49**	97.63±3.51	99.31±1.13	99.11±0.45	99.23±1.25	98.81±0.94	98.22±1.50	99.57±0.63
	C15	88.95±4.43	88.94±6.51	**92.95±6.27**	81.36±8.32	89.21±4.10	89.43±4.34	92.17±6.11	87.66±4.35	92.76±6.71
	C16	99.93±0.11	99.33±0.24	99.44±0.41	98.93±0.66	98.62±3.12	99.48±1.19	98.10±2.66	98.84±1.82	**100.00±0.00**
	OA(%)	**95.30±1.23**	94.28±1.95	95.03±1.02	92.11±0.97	94.26±1.06	92.90±0.99	92.02±1.21	94.06±1.27	94.51±1.01
	AA(%)	97.50±0.61	97.61±0.73	97.60±0.56	96.56±0.44	97.38±0.50	96.61±0.41	95.60±0.79	96.96±0.56	**97.69±0.54**
	κ	**0.9477±0.014**	0.9364±0.022	0.9448±0.011	0.9124±0.011	0.9363±0.012	0.9212±0.011	0.9114±0.013	0.9340±0.014	0.9391±0.011

is relatively straightforward for learning, and even though the learning task of MCNN is simple, it can still learn good representations. Probably, when facing a simple dataset, MAE could try to adjust the masking ratio in the hope of learning better representations. Rather than adjusting the masking ratio or other hyper-parameters, RMAE improves MAE in a way that increases the difficulty of the learning task, and from the experimental results, RMAE improved OA, AA, and κ by 2.49%, 2.09%, and 0.0277, respectively, compared to MAE.

Next, by using a random seed (42), we trained or tuned all models with the same labeled samples and obtained the classification maps shown in Fig. 12. We can find that although A2S2K-RN obtained the highest accuracy, the classification map obtained by RMAE contains less noise, achieving a better visual performance. In addition, we can see that the classification map obtained by RMAE has a regular boundary texture.

At last, we combine $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ to evaluate the performance of MCNN, MAE, IMAE, and RMAE. As the data given in Table X, although MCNN achieved a quite small $\bar{d}_{\text{intra}}$, it also obtained an extremely small $\bar{d}_{\text{inter}}$. By comparing MAE and RMAE, we can see that $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ obtained by RMAE are 0.5139 and 1.0632 smaller than those obtained by MAE, respectively. On the whole, the $\bar{d}_{\text{inter}}$ obtained by MCNN, MAE, and RMAE is more than twice the $\bar{d}_{\text{intra}}$ obtained by each of them, indicating that all of them have excellent tuning performance. In this case, we cannot directly determine who has better tuning performance, but from the experimental results, RMAE achieved the best performance, once again verifying its effectiveness and superiority. For IMAE, the value of $\bar{d}_{\text{inter}}$ it obtained is not greater than twice of $\bar{d}_{\text{intra}}$, which is regarded by us as one of the principal reasons for its mediocre performance on the Salinas dataset. Finally, to better compare MAE and RMAE, we present the 2-D t-SNE embeddings of the representations learned by MAE and RMAE in Fig. 13.

TABLE X
MEAN INTRACLASS AND INTERCLASS DISTANCE OF THE REPRESENTATIONS LEARNED BY MCNN, MAE, IMAE, AND RMAE ON THE SALINAS DATASET

Method	$\bar{d}_{intra}$	$\bar{d}_{inter}$	$\bar{\text{OA}}$(%)	$\bar{\text{AA}}$(%)	$\bar{\kappa}$
MCNN	1.1332	2.4413	92.90	96.61	0.9212
MAE	4.6318	10.8953	92.02	95.60	0.9114
IMAE	8.9405	17.0757	94.06	96.96	0.9340
RMAE	4.1179	9.8321	94.51	97.69	0.9391

V. DISCUSSION

To further understand the effectiveness and advantages of RMAE, we will present more data to discuss the complexity and sensitivity of RMAE. All the analysis experiments were performed on the Salinas dataset as this dataset is one of the most classical datasets in the remote sensing community and has an appropriate proportion of labeled pixels.

A. Complexity

For different datasets, the sample format processed by each model is fixed, so it is necessary to introduce the sample composition of the Salinas dataset in detail before giving the complexity analysis. There are 160 training (tuning) samples, 1600 validation samples, and 53 969 testing samples. For semi-supervised models, they used randomly selected half of the samples, 27 064 samples, as pretraining samples.

The complexity metrics we analyze include the number of trainable parameters, floating-point operations (FLOPs),[4] and GPU memory usage. In addition, for supervised models, we give training time and testing time, and for semi-supervised models, we give the number of tuning parameters, pretraining

[4]For semi-supervised models, FLOPs here only refer to the FLOPs in the pretraining stage.

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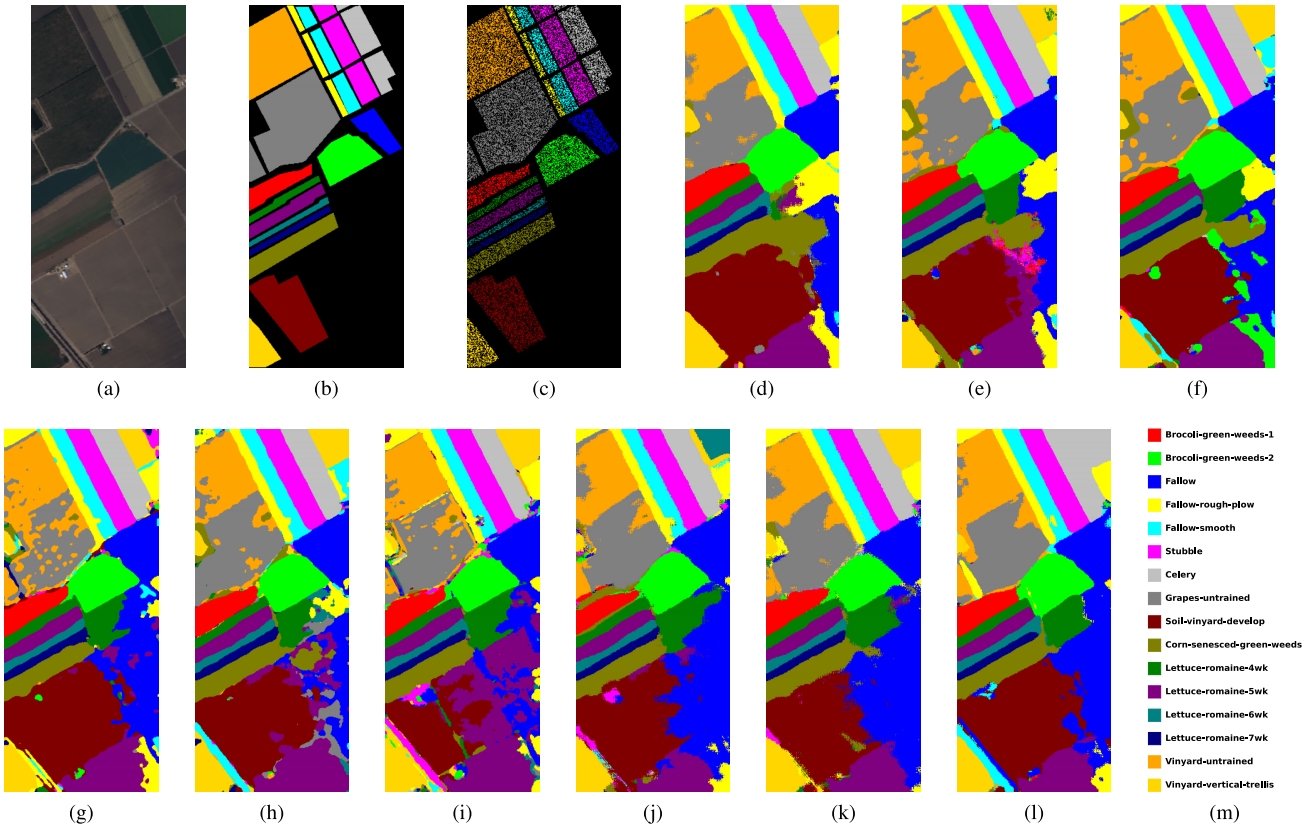


Fig. 12. Visualization of the classification maps obtained by different models on the Salinas dataset. (a) Three-band false-color composite. (b) Ground truth. (c) Pretraining samples. (d)–(l) Classification maps obtained by A2S2K-RN, DMCN, MASSFormer, CapsViT, SSFTT, MCNN, MAE, IMAE, and RMAE. (m) Legend.

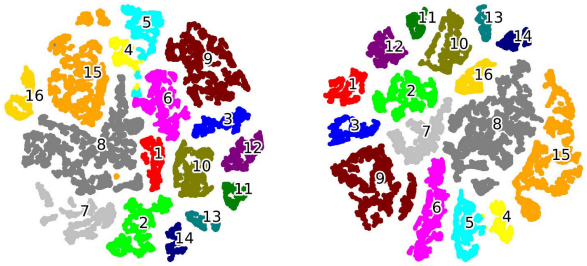


Fig. 13. Two-dimensional t-SNE embeddings of the representations learned by MAE and RMAE on the Salinas dataset. (a) MAE. (b) RMAE.

time, tuning time, and testing time. All statistical time is measured in seconds (s), and all statistical data are presented in Table XI.

As we can see from Table XI, among the supervised models, CapsViT not only has a large number of parameters, but also has high FLOPs. In addition, both DMCN and CapsViT include routing iteration mechanisms, so they require a lot of training time. The lightest model is SSFTT, which not only has a small number of parameters, but also does not occupy a lot of memory, and the most important thing is that it takes the least training time and testing time. From the perspective of trainable parameters and FLOPs, MASSFormer is also a relatively lightweight model, and what is commendable is that its performance is better than SSFTT.

Among the semi-supervised models, despite the fact that IMAE possesses a substantial number of trainable parameters, as it employs a small quantity of ViT blocks in the encoder and decoder (4 and 2, respectively), it demands the least pretraining time. Meanwhile, it also consumes the least GPU memory. Due to the fact that MCNN possesses fewer trainable parameters, it is also capable of completing pretraining rapidly. Of course, we are most concerned about the contrast between MAE and RMAE. We have introduced our self-designed induced transformer into MAE, which uses a small number of visible patches to learn the representations of patches with larger receptive fields, so as to further improve the learning ability of MAE. From the data given in Table XI, we can see that compared with MAE, RMAE has little increase in parameters, and the FLOPs between the two are not significantly different. The memory usage is almost the same. On top of that, for the same 400 epochs, RMAE's pretraining time was only 264 s longer than MAE's. Therefore, we say that RMAE greatly improves the performance of MAE with a small increase in complexity. Finally, with regard to the highly efficient pretraining of IMAE, in practical applications, similar to IMAE, we can adjust the embedding dimension of the encoder (and the induced token) in RMAE to 256, while using fewer ViT blocks in the encoder and decoder to effectively reduce the complexity of RMAE. The reason why

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TABLE XI
COMPARISON OF THE COMPUTATIONAL COSTS OF DIFFERENT MODELS ON THE SALINAS DATASET

Models	Trainable Parameters	Tuning Parameters	FLOPs(10^6)	GPU Memory(MB)	Pre-Training(s)	Training/Tuning(s)	Testing(s)
A2S2K-RN	109843	–	72.246	2454	–	17.39	3.31
DMCN	429760	–	9.455	22798	–	285.96	63.70
MASSFormer	246128	–	29.813	2037	–	27.30	2.71
CapsViT	1973921	–	33.278	8286	–	144.34	31.96
SSFTT	153224	–	16.907	1878	–	8.51	1.18
MCNN	1299823	207568	34.094	1980	1893.99	20.21	1.77
MAE	2814151	207568	27.655	2080	3932.48	20.10	2.59
IMAE	3321238	807248	29.149	526	1546.69	10.99	2.28
RMAE	3028478	207568	29.543	2086	4196.98	31.87	2.71

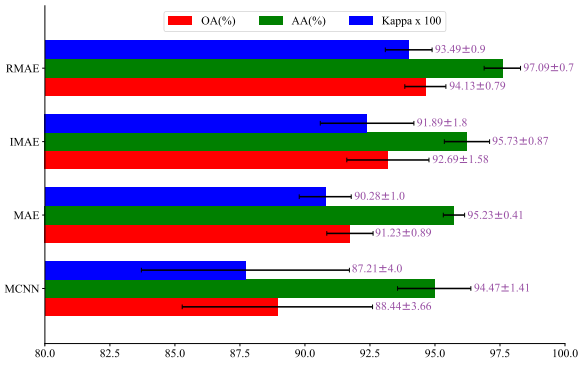


Fig. 14. Linear fine-tuning results obtained by MCNN, MAE, IMAE, and RMAE on the Salinas dataset.

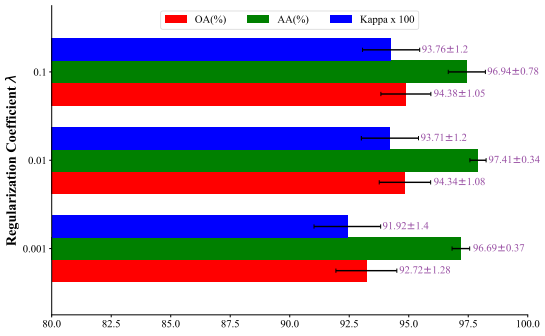


Fig. 15. Effects of different λ on RMAE on the Salinas dataset.

RMAE sets the embedding dimension of the encoder to 128, while using more ViT blocks in the encoder and decoder is to preserve its deep nature to learn more complex, holistic representations.

B. Linear Fine-Tuning

For an SSRL model, the advantage of top-layer unfrozen fine-tuning is that it not only allows the model to retain a large amount of learned knowledge, but also allows the model to flexibly adapt to new samples. In the previous experiments, we had been using top-layer unfrozen fine-tuning for MCNN, MAE, and RMAE. Now, let's combine $\bar{d}_{\text{intra}}$ and $\bar{d}_{\text{inter}}$ to check their linear fine-tuning performance.

For the sake of facilitating comparison, we present the linear fine-tuning results of MCNN, MAE, IMAE, and RMAE in the form of a bar chart in Fig. 14. From the figure, we can see that when we froze all the weights of the encoders in MCNN, MAE, IMAE, and RMAE, and used only two linear layers to perform linear tuning, RMAE still achieved the best classification results. The reason why RMAE can perform very well in linear tuning is because the representations it learned have good separability (see Fig. 13). It is worth noting that MCNN has poor linear tuning performance, and in our experiments, we also observed that it is difficult to fit training samples. Although MCNN achieved slightly better top-layer unfrozen fine-tuning performance than MAE on the Salinas dataset, its linear fine-tuning performance was significantly poor due to the small $\bar{d}_{\text{inter}}$ of the representations it learned.

Finally, by comparing the results given in Fig. 14 with those given in Table IX, we can find that the linear fine-tuning performance of MAE, IMAE, and RMAE all exhibit a minor decline in comparison to their top-layer unfrozen fine-tuning performance. Therefore, in practical applications, we also advocate top-layer unfrozen fine-tuning.

C. Effects of the Regularization Coefficient

RMAE has two reconstruction losses, induced reconstruction loss L_{ir} and masked reconstruction loss L_{mr}, among which L_{ir} is also known as regularization loss. L_{ir} has a regularization coefficient λ that is used to balance the importance between L_{ir} and L_{mr}. In the default configuration, λ is set to 1 because we treat L_{ir} as equally important as L_{mr}. Perhaps the default setting is not the best, for example, in experiments on the MNIST dataset, we set λ to 0.1. Now, let's examine the effects of different λ on RMAE performance and make our final recommendations.

We examine the effects of λ on three scales, namely, 0.001, 0.01, and 0.1. The experimental results are presented in the form of a bar chart in Fig. 15. As can be seen from the picture, when λ was set to 0.001, the performance of RMAE was very close to that of MAE. This is because the regularization constraint of L_{ir} is too small, and the underlying MAE plays a decisive role in the model. When λ was set to 0.01, RMAE began to show its advantages. When λ continued to increase, that is, when it was set to 0.1 or 1, the improvement in RMAE's performance would become weak. But overall, RMAE achieved the best performance when λ was set to 1 (see Table IX).

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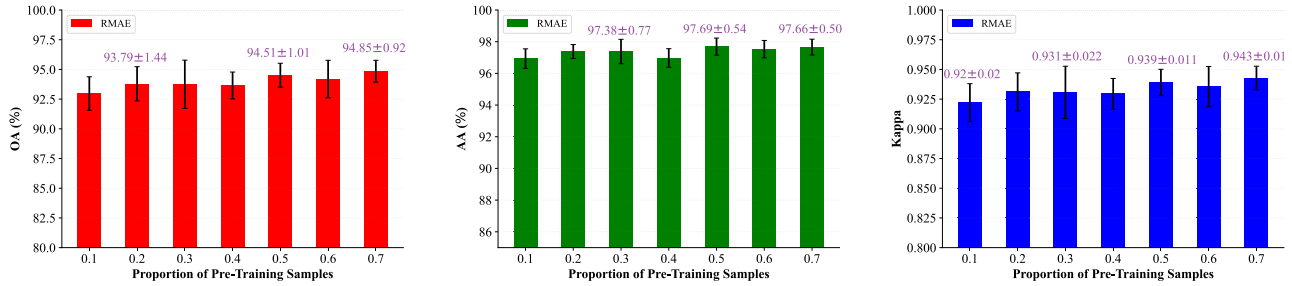


Fig. 16. Performance of RMAE with different proportions of pretrained samples on the Salinas dataset. (a) OA. (b) AA. (c) Kappa.

From the above analysis, we can see that although RMAE is not overly sensitive to λ, from the practical application, HSIs are different from natural images, and it is more difficult to learn, therefore, we recommend setting λ to 1, adding a large regularization constraint, and considering L_{ir} and L_{mr} as equally important.

D. Effects of Different Patch Sizes

In the default configuration of RMAE, the smaller patch size (receptive field) k_1 and larger patch size k_2 are set to 3 and 5, respectively. Now, let's investigate the effects of different patch sizes on RMAE. Specifically, we set the smaller patch size to 1 and investigated the performance of RMAE with larger patch sizes of 3 and 5, respectively.

To better observe the influence of different patch sizes on RMAE, we give the comparison results in Table XII, among them, the pretraining time is given in hours (h). From the data presented, we can see that when k_1 was set to 1, the pretraining time of the entire model increased dramatically. This is because the number of visible patches to be processed increases, leading to an increase in the number of embeddings that the encoder, induced transformer, and decoder need to process. In addition, from the classification results, when k_1 was set to 1, no matter whether k_2 was set to 3 or 5, the performance of RMAE experienced a serious decline. We speculate that there are two reasons for this result: 1) although the number of preserved pixels is almost the same, the density of information received by the encoder increases, leading to a decrease in the difficulty of the learning task; and 2) the task of the decoder has shifted from reconstructing patches to reconstructing pixels, simplifying the reconstruction task and resulting in a decrease in the semantic level of the learned representations. Finally, it is worth mentioning that due to the varying difficulty of the induced transformer's reconstruction task, setting k_2 to 5 is better than 3.

In summary, in practical applications, we should not set the smaller patch size k_1 too small, which can lead to a sharp increase in pretraining time and a decrease in learning ability.

E. Effects of the Proportion of Pretraining Samples

We know that the larger the number of pretraining samples, the more pretraining time the model takes. For the Salinas dataset, we used 0.5 of randomly selected samples as pretraining samples. Now, we investigate the influence of different proportions of pretraining samples on RMAE. Specifically, we randomly select samples ranging from 0.1 to 0.7 as pretraining samples and observe the performance changes of RMAE. The statistics are given in Fig. 16.

TABLE XII
EFFECTS OF DIFFERENT PATCH SIZES ON RMAE ON THE SALINAS DATASET

k_1	k_2	Pre-Training(h)	OA(%)	AA(%)	κ
1	3	3.87	89.01±2.75	95.15±1.04	0.8784±0.030
1	5	3.85	89.42±1.01	95.01±0.63	0.8829±0.011
3	5	1.17	94.51±1.01	97.69±0.54	0.9391±0.011

From Fig. 16, we can see that RMAE did not perform well when the proportion of pretraining samples was 0.1, and the average OA it obtained was less than 93% (92.97%). This is because the number of pretraining samples at this time is too small, which prevents RMAE from learning too much knowledge. When the proportion increased to 0.2, the performance of RMAE was significantly improved. When the proportion came to 0.5, which was the proportion we used in previous experiments, RMAE achieved satisfactory results. RMAE achieved better results when the proportion of pretraining samples continued to increase to 0.7, at which point it obtained an average OA, AA, and κ of 94.85%, 97.66%, and 0.9427, respectively. It is worth noting that, when the proportion was 0.7, the number of pretraining samples reached 37 890, and the pretraining time spent by RMAE was 6047.28 s. Obviously, it took a lot of time.

In practice, if you have sufficient computing resources and time, we suggest setting the proportion of pretraining samples higher so that RMAE can learn better patterns. If you have limited resources to work with, do not be too pessimistic, as RMAE can achieve good performance even with a low percentage of pretraining samples (e.g., 0.2).

F. Effects of the Number of Fine-Tuning Samples

To observe the sensitivity of RMAE to the number of fine-tuning samples, we set the number of fine-tuning samples to 5–15 and observe the performance changes of RMAE. The experimental results obtained are shown in Fig. 17.[5]

[5]For the sake of observation, we only provide the average of the metrics.

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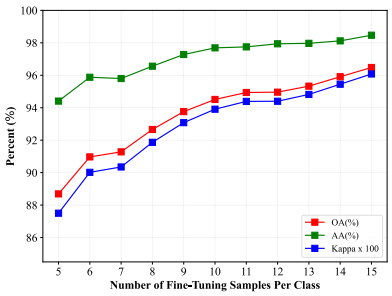


Fig. 17. Effects of the number of fine-tuning samples on RMAE on the Salinas dataset.

From the graph, it can be clearly seen that the performance of RMAE shows a steady upward trend with the increase in the number of fine-tuning samples, which fully proves that RMAE not only has excellent SSRL capability, but also has good tunability. In summary, although RMAE does not perform as well as A2S2K-RN on the Salinas dataset, its biggest advantage is that it can learn perfect representations from a large number of unlabeled samples, providing support for downstream tasks, which is an ability that supervised models do not have.

VI. Conclusion

In this article, we have proposed RMAE to further improve the performance of SSRL-based SHIC. In RMAE, we introduce a self-designed induced transformer block and let it learn the representations of patches with larger receptive fields through a small number of visible patches. By this way, a more challenging self-supervised task is created, enabling RMAE to have a more robust learning ability. Most importantly, the induced transformer block we designed has a similar structure to a single transformer block and does not introduce much computational overhead to the underlying MAE. In addition, by viewing the masking operation as a special pooling, we have developed a simple MCNN to delve into the principles behind the tremendous success of MAE and confirm the effectiveness of RMAE. We hope this perspective will inspire future work. Finally, we give two new metrics, namely, the mean intraclass distance and the mean interclass distance. Based on the two metrics given, we present two criteria for evaluating the tunability of SSRL models. Experiments on four publicly available HSI datasets demonstrate the effectiveness and superiority of RMAE. Especially for the HanChuan dataset, RMAE achieved an average OA of 90.04% using only one-fifth of the pretraining samples and ten fine-tuning samples per class.

A minor limitation of RMAE is that it demands the spatial size of the sample to be evenly divided by both the smaller and larger patch sizes. However, as mentioned in the experimental setup, the spatial size of the sample is overly large, leading to numerous redundant calculations, while being too small would constrain the contained spatial information. Hence, we suggest the default configuration of RMAE. In the future, how to explicitly construct RMAE learning representations with a smaller mean intraclass distance and a larger mean interclass distance is our focus. What's more, it will constitute a highly intriguing research topic to investigate the application of RMAE in unsupervised HIC.

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